

# INTERNATIONAL STANDARD



---

**Universal serial bus interfaces for data and power –  
Part 1-4: Common components – USB Type-C™ Authentication Specification**



## THIS PUBLICATION IS COPYRIGHT PROTECTED

Copyright © 2018 IEC, Geneva, Switzerland

All rights reserved. Unless otherwise specified, no part of this publication may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm, without permission in writing from either IEC or IEC's member National Committee in the country of the requester. If you have any questions about IEC copyright or have an enquiry about obtaining additional rights to this publication, please contact the address below or your local IEC member National Committee for further information.

IEC Central Office  
3, rue de Varembe  
CH-1211 Geneva 20  
Switzerland

Tel.: +41 22 919 02 11  
[info@iec.ch](mailto:info@iec.ch)  
[www.iec.ch](http://www.iec.ch)

### About the IEC

The International Electrotechnical Commission (IEC) is the leading global organization that prepares and publishes International Standards for all electrical, electronic and related technologies.

### About IEC publications

The technical content of IEC publications is kept under constant review by the IEC. Please make sure that you have the latest edition, a corrigenda or an amendment might have been published.

#### IEC Catalogue - [webstore.iec.ch/catalogue](http://webstore.iec.ch/catalogue)

The stand-alone application for consulting the entire bibliographical information on IEC International Standards, Technical Specifications, Technical Reports and other documents. Available for PC, Mac OS, Android Tablets and iPad.

#### IEC publications search - [webstore.iec.ch/advsearchform](http://webstore.iec.ch/advsearchform)

The advanced search enables to find IEC publications by a variety of criteria (reference number, text, technical committee,...). It also gives information on projects, replaced and withdrawn publications.

#### IEC Just Published - [webstore.iec.ch/justpublished](http://webstore.iec.ch/justpublished)

Stay up to date on all new IEC publications. Just Published details all new publications released. Available online and also once a month by email.

#### Electropedia - [www.electropedia.org](http://www.electropedia.org)

The world's leading online dictionary of electronic and electrical terms containing 21 000 terms and definitions in English and French, with equivalent terms in 16 additional languages. Also known as the International Electrotechnical Vocabulary (IEV) online.

#### IEC Glossary - [std.iec.ch/glossary](http://std.iec.ch/glossary)

67 000 electrotechnical terminology entries in English and French extracted from the Terms and Definitions clause of IEC publications issued since 2002. Some entries have been collected from earlier publications of IEC TC 37, 77, 86 and CISPR.

#### IEC Customer Service Centre - [webstore.iec.ch/csc](http://webstore.iec.ch/csc)

If you wish to give us your feedback on this publication or need further assistance, please contact the Customer Service Centre: [sales@iec.ch](mailto:sales@iec.ch).



IEC 62680-1-4

Edition 1.0 2018-04

# INTERNATIONAL STANDARD



---

**Universal serial bus interfaces for data and power –  
Part 1-4: Common components – USB Type-C™ Authentication Specification**

INTERNATIONAL  
ELECTROTECHNICAL  
COMMISSION

ICS 35.200

ISBN 978-2-8322-5533-9

**Warning! Make sure that you obtained this publication from an authorized distributor.**

## INTERNATIONAL ELECTROTECHNICAL COMMISSION

## UNIVERSAL SERIAL BUS INTERFACES FOR DATA AND POWER –

## Part 1-4: Common components – USB Type-C™ Authentication Specification

## FOREWORD

- 1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as “IEC Publication(s)”). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
- 2) The formal decisions or agreements of IEC on technical matters express, as nearly as possible, an international consensus of opinion on the relevant subjects since each technical committee has representation from all interested IEC National Committees.
- 3) IEC Publications have the form of recommendations for international use and are accepted by IEC National Committees in that sense. While all reasonable efforts are made to ensure that the technical content of IEC Publications is accurate, IEC cannot be held responsible for the way in which they are used or for any misinterpretation by any end user.
- 4) In order to promote international uniformity, IEC National Committees undertake to apply IEC Publications transparently to the maximum extent possible in their national and regional publications. Any divergence between any IEC Publication and the corresponding national or regional publication shall be clearly indicated in the latter.
- 5) IEC itself does not provide any attestation of conformity. Independent certification bodies provide conformity assessment services and, in some areas, access to IEC marks of conformity. IEC is not responsible for any services carried out by independent certification bodies.
- 6) All users should ensure that they have the latest edition of this publication.
- 7) No liability shall attach to IEC or its directors, employees, servants or agents including individual experts and members of its technical committees and IEC National Committees for any personal injury, property damage or other damage of any nature whatsoever, whether direct or indirect, or for costs (including legal fees) and expenses arising out of the publication, use of, or reliance upon, this IEC Publication or any other IEC Publications.
- 8) Attention is drawn to the Normative references cited in this publication. Use of the referenced publications is indispensable for the correct application of this publication.
- 9) Attention is drawn to the possibility that some of the elements of this IEC Publication may be the subject of patent rights. IEC shall not be held responsible for identifying any or all such patent rights.

International Standard IEC 62680-1-4 has been prepared by technical area 14: Interfaces and methods of measurement for personal computing equipment, of IEC technical committee 100: Audio, video and multimedia systems and equipment.

The text of this standard was prepared by the USB Implementers Forum (USB-IF). The structure and editorial rules used in this publication reflect the practice of the organization which submitted it.

The text of this International Standard is based on the following documents:

| CDV          | Report on voting |
|--------------|------------------|
| 100/2981/CDV | 100/3046/RVC     |

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

A list of all parts in the IEC 62680 series, published under the general title *Universal serial bus interfaces for data and power*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

**IMPORTANT – The 'colour inside' logo on the cover page of this publication indicates that it contains colours which are considered to be useful for the correct understanding of its contents. Users should therefore print this document using a colour printer.**

## INTRODUCTION

The IEC 62680 series is based on a series of specifications that were originally developed by the USB Implementers Forum (USB-IF). These specifications were submitted to the IEC under the auspices of a special agreement between the IEC and the USB-IF.

This standard is the USB-IF publication USB Type-C™ Authentication Specification Revision 1.0.

The USB Implementers Forum, Inc.(USB-IF) is a non-profit corporation founded by the group of companies that developed the Universal Serial Bus specification. The USB-IF was formed to provide a support organization and forum for the advancement and adoption of Universal Serial Bus technology. The Forum facilitates the development of high-quality compatible USB peripherals (devices), and promotes the benefits of USB and the quality of products that have passed compliance testing.

**ANY USB SPECIFICATIONS ARE PROVIDED TO YOU "AS IS, "WITH NO WARRANTIES WHATSOEVER, INCLUDING ANY WARRANTY OF MERCHANTABILITY, NON-INFRINGEMENT, OR FITNESS FOR ANY PARTICULAR PURPOSE. THE USB IMPLEMENTERS FORUM AND THE AUTHORS OF ANY USB SPECIFICATIONS DISCLAIM ALL LIABILITY, INCLUDING LIABILITY FOR INFRINGEMENT OF ANY PROPRIETARY RIGHTS, RELATING TO USE OR IMPLEMENTATION OR INFORMATION IN THIS SPECIFICATION.**

**THE PROVISION OF ANY USB SPECIFICATIONS TO YOU DOES NOT PROVIDE YOU WITH ANY LICENSE, EXPRESS OR IMPLIED, BY ESTOPPEL OR OTHERWISE, TO ANY INTELLECTUAL PROPERTY RIGHTS.**

Entering into USB Adopters Agreements may, however, allow a signing company to participate in a reciprocal, RAND-Z licensing arrangement for compliant products. For more information, please see:

<http://www.usb.org/developers/docs/>

[http://www.usb.org/developers/devclass\\_docs#approved](http://www.usb.org/developers/devclass_docs#approved)

IEC DOES NOT TAKE ANY POSITION AS TO WHETHER IT IS ADVISABLE FOR YOU TO ENTER INTO ANY USB ADOPTERS AGREEMENTS OR TO PARTICIPATE IN THE USB IMPLEMENTERS FORUM.”

# **Universal Serial Bus Type-C™ Authentication Specification**

**Revision 1.0 with ECN and Errata through February 2, 2017**

Copyright © 2017 USB 3.0 Promoter Group. All rights reserved.

**Copyright © 2017, USB 3.0 Promoter Group  
All rights reserved.**

INTELLECTUAL PROPERTY DISCLAIMER

THIS SPECIFICATION IS PROVIDED TO YOU “AS IS” WITH NO WARRANTIES WHATSOEVER, INCLUDING ANY WARRANTY OF MERCHANTABILITY, NON-INFRINGEMENT, OR FITNESS FOR ANY PARTICULAR PURPOSE. THE AUTHORS OF THIS SPECIFICATION DISCLAIM ALL LIABILITY, INCLUDING LIABILITY FOR INFRINGEMENT OF ANY PROPRIETARY RIGHTS, RELATING TO USE OR IMPLEMENTATION OF INFORMATION IN THIS SPECIFICATION. THE PROVISION OF THIS SPECIFICATION TO YOU DOES NOT PROVIDE YOU WITH ANY LICENSE, EXPRESS OR IMPLIED, BY ESTOPPEL OR OTHERWISE, TO ANY INTELLECTUAL PROPERTY RIGHTS.

All product names are trademarks, registered trademarks, or service marks of their respective owners.

USB Type-C™ and USB-C™ are trademarks of USB Implementers Forum.

Copyright © 2017 USB 3.0 Promoter Group. All rights reserved.



## CONTENTS

|                                                                 |    |
|-----------------------------------------------------------------|----|
| Specification Work Group Chairs / Specification Editors .....   | 12 |
| Specification Work Group Contributors .....                     | 12 |
| Revision History .....                                          | 14 |
| 1 Introduction .....                                            | 15 |
| 1.1 Scope .....                                                 | 15 |
| 1.2 Overview .....                                              | 15 |
| 1.3 Related Documents .....                                     | 16 |
| 1.4 Terms and Abbreviations .....                               | 18 |
| 1.5 Conventions .....                                           | 19 |
| 1.5.1 Precedence .....                                          | 19 |
| 1.5.2 Keywords .....                                            | 19 |
| 1.5.3 Numbering .....                                           | 20 |
| 1.5.4 Byte Ordering .....                                       | 20 |
| 2 Overview .....                                                | 20 |
| 2.1 Topology .....                                              | 20 |
| 2.2 Cryptographic Methods .....                                 | 21 |
| 2.2.1 Random Numbers .....                                      | 21 |
| 2.3 Security Overview .....                                     | 22 |
| 2.3.1 Periodic Re-Authentication .....                          | 22 |
| 2.3.2 Secret Key Storage and Protection .....                   | 22 |
| 2.3.3 Security Evaluation Criteria .....                        | 22 |
| 2.4 Impact to Existing Ecosystem .....                          | 22 |
| 2.4.1 Proxy Capabilities (PD traversing the Hub topology) ..... | 23 |
| 3 Authentication Architecture .....                             | 23 |
| 3.1 Certificates .....                                          | 23 |
| 3.1.1 Format .....                                              | 23 |
| 3.1.2 Textual Format .....                                      | 23 |
| 3.1.3 Attributes and Extensions .....                           | 23 |
| 3.2 Certificate Chains .....                                    | 25 |
| 3.2.1 Provisioning .....                                        | 25 |
| 3.3 Private Keys .....                                          | 26 |
| 4 Authentication Protocol .....                                 | 26 |
| 4.1 Digest Query .....                                          | 26 |
| 4.2 Certificate Chain Read .....                                | 26 |
| 4.3 Authentication Challenge .....                              | 27 |
| 4.4 Errors and Alerts .....                                     | 27 |
| 4.4.1 Invalid Request .....                                     | 27 |
| 4.4.2 Unsupported Protocol Version .....                        | 27 |
| 4.4.3 Busy .....                                                | 27 |

Copyright © 2017 USB 3.0 Promoter Group. All rights reserved.

|       |                                                                |    |
|-------|----------------------------------------------------------------|----|
| 4.4.4 | Unspecified .....                                              | 27 |
| 5     | Authentication Messages .....                                  | 27 |
| 5.1   | Header .....                                                   | 28 |
| 5.1.1 | USB Type-C Authentication Protocol Version .....               | 28 |
| 5.1.2 | Message Type .....                                             | 28 |
| 5.1.3 | Param1 .....                                                   | 28 |
| 5.1.4 | Param2 .....                                                   | 28 |
| 5.2   | Authentication Requests .....                                  | 28 |
| 5.2.1 | GET_DIGESTS .....                                              | 29 |
| 5.2.2 | GET_CERTIFICATE .....                                          | 29 |
| 5.2.3 | CHALLENGE .....                                                | 30 |
| 5.3   | Authentication Responses .....                                 | 30 |
| 5.3.1 | DIGESTS .....                                                  | 31 |
| 5.3.2 | CERTIFICATE .....                                              | 31 |
| 5.3.3 | CHALLENGE_AUTH .....                                           | 32 |
| 5.3.4 | ERROR .....                                                    | 33 |
| 6     | Authentication of PD Products .....                            | 34 |
| 6.1   | Transfers less than or equal to <i>MaxExtendedMsgLen</i> ..... | 34 |
| 6.2   | Transfers greater than <i>MaxExtendedMsgLen</i> .....          | 35 |
| 6.3   | Timing Requirements for PD Security Extended Messages .....    | 38 |
| 6.3.1 | Authentication Initiator .....                                 | 38 |
| 6.3.2 | Authentication Responder .....                                 | 39 |
| 6.4   | Context Hash .....                                             | 40 |
| 7     | Authentication of USB Products .....                           | 40 |
| 7.1   | Descriptors .....                                              | 40 |
| 7.1.1 | Authentication Capability Descriptor .....                     | 40 |
| 7.2   | Mapping Authentication Messages to USB .....                   | 41 |
| 7.2.1 | Authentication IN .....                                        | 41 |
| 7.2.2 | Authentication OUT .....                                       | 42 |
| 7.3   | Authentication Protocol .....                                  | 42 |
| 7.3.1 | Digest Query .....                                             | 42 |
| 7.3.2 | Certificate Read .....                                         | 43 |
| 7.3.3 | Authentication Challenge .....                                 | 43 |
| 7.3.4 | Errors .....                                                   | 44 |
| 7.4   | Timing Requirements for USB .....                              | 44 |
| 7.4.1 | USB Host Timing Requirements .....                             | 44 |
| 7.4.2 | USB Device Timing Requirements .....                           | 45 |
| 7.5   | Context Hash .....                                             | 46 |
| 8     | Protocol Constants .....                                       | 46 |
| A     | ACD .....                                                      | 47 |
| A.1.  | ACD Formatting .....                                           | 47 |

|        |                                                     |    |
|--------|-----------------------------------------------------|----|
| A.1.1. | Version TLV .....                                   | 47 |
| A.1.2. | XID TLV .....                                       | 48 |
| A.1.3. | Power Source Capabilities TLV .....                 | 48 |
| A.1.4. | Power Source Certifications TLV .....               | 49 |
| A.1.5. | Cable Capabilities TLV .....                        | 50 |
| A.1.6. | Security Description TLV .....                      | 50 |
| A.1.7. | Playpen TLV .....                                   | 54 |
| A.1.8. | Vendor Extension TLV .....                          | 55 |
| A.1.9. | Extension TLV .....                                 | 55 |
| A.2.   | ACD for a PD Product .....                          | 55 |
| A.3.   | ACD for a USB Product .....                         | 56 |
| B      | Cryptographic Examples .....                        | 57 |
| B.1.   | Example Authentication Sequence .....               | 57 |
| B.2.   | Example Certificate Chain Topology .....            | 57 |
| B.2.1. | Certificate Chain .....                             | 57 |
| B.2.2. | Root Certificate .....                              | 62 |
| B.2.3. | Key Pairs .....                                     | 63 |
| B.3.   | Example Authentication Signature Verification ..... | 64 |
| B.3.1. | CHALLENGE Request .....                             | 64 |
| B.3.2. | CHALLENGE_AUTH Response .....                       | 64 |
| C      | Potential Attack Vectors .....                      | 65 |

## TABLES

|                                                             |    |
|-------------------------------------------------------------|----|
| Table 1-1: Terms and Abbreviations .....                    | 18 |
| Table 2-1: Summary of Cryptographic Methods .....           | 21 |
| Table 3-1: Certificate Chain Format .....                   | 25 |
| Table 5-1: Authentication Message Header .....              | 28 |
| Table 5-2: USB Type-C Authentication Protocol Version ..... | 28 |
| Table 5-3: Authentication Request Types .....               | 29 |
| Table 5-4: GET_DIGESTS Request Header .....                 | 29 |
| Table 5-5: GET_CERTIFICATE Request Header .....             | 29 |
| Table 5-6: GET_CERTIFICATE Request Payload .....            | 30 |
| Table 5-7: CHALLENGE Request Header .....                   | 30 |
| Table 5-8: CHALLENGE Request Payload .....                  | 30 |
| Table 5-9: Authentication Response Types .....              | 30 |
| Table 5-10: DIGESTS Response Header .....                   | 31 |
| Table 5-11: DIGESTS Response Payload .....                  | 31 |
| Table 5-12: CERTIFICATE Response Header .....               | 31 |
| Table 5-13: CERTIFICATE Response Payload .....              | 32 |

Copyright © 2017 USB 3.0 Promoter Group. All rights reserved.

|                                                                            |    |
|----------------------------------------------------------------------------|----|
| Table 5-14: CHALLENGE_AUTH Response Header .....                           | 32 |
| Table 5-15: CHALLENGE_AUTH Response Payload .....                          | 33 |
| Table 5-16: Message Contents for ECDSA Digital Signature .....             | 33 |
| Table 5-17: ERROR Response Header .....                                    | 34 |
| Table 5-18: ERROR Codes .....                                              | 34 |
| Table 6-1: Timeout Values for a PD Authentication Initiator .....          | 38 |
| Table 6-2: Timing Requirements for PD Authentication Responder .....       | 39 |
| Table 7-1: Authentication Capability Descriptor .....                      | 40 |
| Table 7-2: Authentication Capability Descriptor Types .....                | 41 |
| Table 7-3: Authentication Message <i>bRequest</i> Values .....             | 41 |
| Table 7-4: Authentication IN Control Request Fields .....                  | 41 |
| Table 7-5: Authentication Message Header Mapping .....                     | 41 |
| Table 7-6: Authentication OUT Control Request Fields .....                 | 42 |
| Table 7-7: GET_DIGESTS Authentication IN Control Request Fields .....      | 42 |
| Table 7-8: GET_CERTIFICATE Authentication OUT Control Request Fields ..... | 43 |
| Table 7-9: CERTIFICATE Authentication IN Control Request Fields .....      | 43 |
| Table 7-10: CHALLENGE Authentication OUT Control Request Fields .....      | 43 |
| Table 7-11: CHALLENGE_AUTH Authentication IN Control Request Fields .....  | 44 |
| Table 7-12: Authentication Initiator Timeout Values .....                  | 44 |
| Table 7-13: Authentication Responder Response Times .....                  | 45 |
| Table 8-1: Protocol Constants .....                                        | 46 |
| Table A-1: TLV General Format .....                                        | 47 |
| Table A-2: TLV Types .....                                                 | 47 |
| Table A-3: Version TLV Fields .....                                        | 47 |
| Table A-4: ACD Version Encoding .....                                      | 48 |
| Table A-5: XID TLV Fields .....                                            | 48 |
| Table A-6: Power Source Capabilities TLV Fields .....                      | 48 |
| Table A-7: Power Source Capabilities TLV Data .....                        | 49 |
| Table A-8: Power Source Certifications TLV Fields .....                    | 49 |
| Table A-9: Cable Capabilities TLV Fields .....                             | 50 |
| Table A-10: Cable Capabilities TLV Data .....                              | 50 |
| Table A-11: Security Description TLV Fields .....                          | 50 |
| Table A-12: Security Data .....                                            | 50 |
| Table A-13: FIPS/ISO Level Identifiers .....                               | 51 |
| Table A-14: Vulnerability Assessment .....                                 | 51 |
| Table A-15: EAL Encodings .....                                            | 52 |
| Table A-16: Protection Profile Encoding .....                              | 52 |

|                                                       |    |
|-------------------------------------------------------|----|
| Table A-17: Development Security .....                | 53 |
| Table A-18: Certification Maintenance .....           | 53 |
| Table A-19: Testing Method Encoding .....             | 54 |
| Table A-20: Vulnerability Assessment .....            | 54 |
| Table A-21: Playpen TLV Fields .....                  | 55 |
| Table A-22: Vendor Extension TLV Fields .....         | 55 |
| Table A-23: Vendor Extension TLV Data .....           | 55 |
| Table A-24: Extension TLV Fields .....                | 55 |
| Table A-25: PD Product ACD TLVs .....                 | 56 |
| Table A-26: USB Product ACD TLVs .....                | 56 |
| Table B-1: Version TLV Fields .....                   | 61 |
| Table B-2: XID TLV Fields .....                       | 61 |
| Table B-3: Power Source Capabilities TLV Fields ..... | 61 |
| Table B-4: Security Description TLV Fields .....      | 61 |
| Table B-5: Playpen TLV Fields .....                   | 62 |
| Table B-6: Vendor Extension TLV Fields .....          | 62 |

## FIGURES

|                                                                                    |    |
|------------------------------------------------------------------------------------|----|
| Figure 2-1 Sample Topology .....                                                   | 21 |
| Figure 6-1 Example Security Transfer Process for an Authentication Initiator ..... | 36 |
| Figure 6-2 Example Security Transfer Process for an Authentication Responder ..... | 37 |
| Figure 6-3 Example 612-Byte Certificate Chain Read .....                           | 38 |
| Figure A-1: Bitmap of Version TLV Data .....                                       | 48 |
| Figure A-2: Bitmap of the Common Criteria Identifier .....                         | 51 |
| Figure A-3: Bitmap of the Security Analysis Identifier .....                       | 53 |

## Specification Work Group Chairs / Specification Editors

|                           |          |                   |
|---------------------------|----------|-------------------|
| Renesas Electronics Corp. | Co-Chair | Bob Dunstan       |
| Intel Corporation         | Co-Chair | Abdul Ismail      |
|                           | Editor   | Stephanie Wallick |

## Specification Work Group Contributors

|                           |                          |                    |                     |
|---------------------------|--------------------------|--------------------|---------------------|
| Advanced Micro Devices    | Jason Hawken             | Joseph Scanlon     |                     |
| Apple                     | Colin Whitby-Strevens    | Robert Walsh       | Reese Schreiber     |
|                           | David Conroy             | David Sekowski     |                     |
| Atmel Corporation         | Kerry Maletsky           | Stephen Clark      | Michel Guellec      |
|                           | Ronald Ih                |                    |                     |
| Cypress Semiconductor     | Subu Sankaran            | Jagadeesan Raj     | Anup Nayak          |
|                           | Jan-Willem van der Waert |                    |                     |
| Dell Inc.                 | Sean O'Neal              | Mohammed Hijazi    | Frank Molsberry     |
|                           | Dan Hamlin               | Rick Martinez      |                     |
| DisplayLink (UK) Ltd.     | Richard Petrie           | Pete Burgers       | Dan Ellis           |
| Fresco Logic Inc.         | Bob McVay                | Tom Burton         | Christopher Meyers  |
|                           | Thomas Huang             |                    |                     |
| Google Inc.               | Adam Langley             | William Richardson | Adam Rodriguez      |
|                           | David Schneider          | Mark Hayter        | Ken Wu              |
|                           | Will Drewry              | Jerry Parson       | Sanjay Krishnan     |
| HP Inc.                   | Alan Berkema             | Jim Waldron        | Daniel Hong         |
| Infineon Technologies     | Thomas Poeppelmann       | Wolfgang Furtner   | Harald Hewel        |
|                           | Wieland Fischer          | Sie Boo Chiang     |                     |
| Intel Corporation         | Brad Saunders            | David Johnston     | Chia-Hung Kuo       |
|                           | Christine Krause         | Rolf Kuhn          | Steve McGowan       |
|                           | Andrew Reinders          | Purushottam Goel   | Karthi Vadivelu     |
| Lattice Semiconductor     | Hoon Choi                | Thomas Watzka      |                     |
| MCCI Corporation          | Terry Moore              |                    |                     |
| Microchip Technology Inc. | Richard Wahler           | Mark Bohm          | Atish Ghosh         |
|                           | Robert Schoepflin        |                    |                     |
| Microsoft Corporation     | Niels Ferguson           | Nathan Sherman     | Martin Borge        |
|                           | Kinshumann Kinshumann    | Vivek Gupta        | Toby Nixon          |
|                           | Kai Inha                 | Robbie Harris      | Andrea Keating      |
|                           | Fred Bhesania            | Jayson Kastens     | Rahul Ramadas       |
| NXP Semiconductors        | Vijendra Kuroodi         | Joe Salvador       | Alicia da Conceição |
|                           | Krishnan TN              |                    |                     |
| Renesas Electronics Corp. | Philip Leung             | Hideyuki Tanaka    | Yuji Asano          |
|                           | Kentaro Omata            | Yoshiyuki Tomoda   | Kiichi Muto         |

Copyright © 2017 USB 3.0 Promoter Group. All rights reserved.

|                               |                        |                     |                   |
|-------------------------------|------------------------|---------------------|-------------------|
| ROHM Co., Ltd.                | Masahiko Nagata        | Chizuru Matsunaga   | Toshifumi Yamaoka |
|                               | Ruben Balbuena         | Kris Bahar          | Nobutaka Itakura  |
|                               | Takashi Sato           |                     |                   |
| Samsung Electronics Co., Ltd. | Tong Kim               | Jagoun Koo          | Soondo Kim        |
| STMicroelectronics            | Enrico Gregoratto      | Guido Bertoni       | Sylvie Wuidart    |
|                               | Yannick Teglia         | Anis Ben-Abdallah   | Massimo Panzica   |
|                               | Andrew Marsh           | Joris Delclef       | Nathalie Ballot   |
|                               | Joel Huloux            | Bernard Kasser      | Dragos Davidescu  |
|                               | Christophe Lorin       |                     |                   |
| Synopsys, Inc.                | Eric Huang             | Morten Christiansen | Gervais Fong      |
|                               | Venkataraman Krishnan  | Nivin George        | Aaron Yang        |
|                               | Subramaniam Aravindhan | Bala Babu           | Satya Patnala     |
|                               | Kevin Heilman          | John Youn           | Zongyao Wen       |
| Texas Instruments             | Charles Campbell       | Deric Waters        | Scott Jackson     |
| Total Phase                   | Chris Yokum            |                     |                   |
| VIA Technologies              | Terrance Shih          | Jay Tseng           | Fong-Jim Wang     |
|                               | Benjamin Pan           |                     |                   |

**Revision History**

| <b>Revision</b>      | <b>Date</b>      | <b>Description</b>                               |
|----------------------|------------------|--------------------------------------------------|
| 1.0                  | March 25, 2016   | Initial Release                                  |
| 1.0 + ECN and Errata | February 2, 2017 | Includes ECN and errata through February 2, 2017 |

Copyright © 2017 USB 3.0 Promoter Group. All rights reserved.



## 1 Introduction

This specification provides a means for authenticating Products with regard to identification and configuration. Authentication is performed via USB Power Delivery message communications and/or via USB data bus control transactions.

USB Type-C™ Authentication allows an organization to set and enforce a Policy with regard to acceptable Products. This will permit useful security assurances in real world situations. For example:

- A vendor, concerned about product damage resulting from substandard charging devices, can set a Policy requiring that only certified PD Products be used for charging.
- A user, concerned about charging his phone at a public terminal, can set a Policy in his phone requiring that the phone only charge from certified PD Products.
- An organization, concerned about unidentifiable storage devices gaining access to corporate PC assets, can set a Policy in its PCs requiring that only USB storage devices that have been verified and signed by corporate IT are used.

### 1.1 Scope

This specification defines the architecture and methodology for unilateral Product Authentication. It is intended to be fully compatible with and extend existing PD and USB infrastructure. Information is provided to allow for Policy enforcement, but individual Policy decisions are not specified.

The Authentication of USB Type-C products that support Alternate Modes is allowed. However, the methods to do so are outside the scope of this specification.

### 1.2 Overview

This specification provides primitives for unilateral Authentication. The security model defined by this specification permits assurances that a Product is:

- Of a particular type from a particular manufacturer with particular characteristics
- Owned and controlled by a particular organization

Local Policy will determine which features need to be present in an attached Product before accessing or providing a resource (e.g. power, storage, etc.).

Product vendors can add security features beyond those listed in this specification, but the definition and implementation of those features is up to the vendor. Added features cannot alter the base specifications defined herein.

### 1.3 Related Documents

- **USB2.0** – Universal Serial Bus Specification, Revision 2.0, (including errata and ECNs through August 11, 2014) (referred to in this document as the USB 2.0 Specification) (available at: <http://www.usb.org/developers/docs>.)
- **USB3.1** – Universal Serial Bus 3.1 Specification, Revision 1.0, (including errata and ECNs through August 11, 2014) (referred to in this document as the USB 3.1 Specification) (available at: <http://www.usb.org/developers/docs>.)
- **USBPD** – Universal Serial Bus Power Delivery Specification, Revision 3, Version 1.0a, March 25, 2016 (referred to in this document as the USB PD Specification) (available at: <http://www.usb.org/developers/docs>.)
- **USBTYPEC** – Universal Serial Bus Type-C Cable and Connector Specification, Revision 1.2, March 25, 2016 (referred to in this document as the USB Type-C Specification)(available at: <http://www.usb.org/developers/docs>.)
- **USBTYPEC BRIDGE** Universal Serial Bus Type-C Bridge Specification, Revision 1.0, March 25, 2016, (available at <http://www.usb.org/developers/docs>.)
- **ASN.1** - ISO-822-1-4;
  - ITU-T X.680 (available at: [https://www.itu.int/rec/dologin\\_pub.asp?lang=e&id=T-REC-X.680-201508-!!!PDF-E&type=items](https://www.itu.int/rec/dologin_pub.asp?lang=e&id=T-REC-X.680-201508-!!!PDF-E&type=items));
  - ITU-T X.681 (available at: [https://www.itu.int/rec/dologin\\_pub.asp?lang=e&id=T-REC-X.681-201508-!!!PDF-E&type=items](https://www.itu.int/rec/dologin_pub.asp?lang=e&id=T-REC-X.681-201508-!!!PDF-E&type=items));
  - ITU-T X.682 (Available at: [https://www.itu.int/rec/dologin\\_pub.asp?lang=e&id=T-REC-X.682-201508-!!!PDF-E&type=items](https://www.itu.int/rec/dologin_pub.asp?lang=e&id=T-REC-X.682-201508-!!!PDF-E&type=items));
  - ITU-T X.683 (Available at: [https://www.itu.int/rec/dologin\\_pub.asp?lang=e&id=T-REC-X.683-201508-!!!PDF-E&type=items](https://www.itu.int/rec/dologin_pub.asp?lang=e&id=T-REC-X.683-201508-!!!PDF-E&type=items).)
- **DER** - ISO-8825-1; ITU-T X.690 (available at: [https://www.itu.int/rec/dologin\\_pub.asp?lang=e&id=T-REC-X.690-201508-!!!PDF-E&type=items](https://www.itu.int/rec/dologin_pub.asp?lang=e&id=T-REC-X.690-201508-!!!PDF-E&type=items).)
- **X509v3** - ISO-9594-8; ITU-T X.509 (available at: [https://www.itu.int/rec/dologin\\_pub.asp?lang=e&id=T-REC-X.509-201210-!!!PDF-E&type=items](https://www.itu.int/rec/dologin_pub.asp?lang=e&id=T-REC-X.509-201210-!!!PDF-E&type=items).)
- **Common Criteria**:
  - Common Criteria for Information Technology Security Evaluation, Parts 1-3, Version 3.1, Revision 4, September 2010 (available at: <https://www.commoncriteriaportal.org/cc/#supporting> )
  - ISO/IEC 15408 Evaluation criteria for IT security Parts 1-3
- **ECDSA**:
  - ANSI X9.62; NIST-FIPS-186-4, Section 6 (available at: <http://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.186-4.pdf>.)
  - ISO/IEC 14888-3 Digital signatures with appendix -- Part 3: Discrete logarithm based mechanisms (Clause 6.6)
- **NIST P256, secp256r1**:
  - Certicom-SEC-2 (available at: <http://www.secg.org/sec2-v2.pdf>); NIST-Recommended-EC (available at: <http://csrc.nist.gov/groups/ST/toolkit/documents/dss/NISTReCur.pdf>.)

Copyright © 2017 USB 3.0 Promoter Group. All rights reserved.

- ISO/IEC 15946 Cryptographic techniques based on elliptic curves (NIST P-256 is included as example)
  - *Notes: ISO/IEC 15946 series treat elliptic curves differently from FIPS 186-4. ISO/IEC 15946-5 is about elliptic curve generation. That is, based on the method in part 5, each application and implementation can generate its own curves to use. In other words, no ISO/IEC recommended curves. P-256 is consider an example in ISO/IEC 15946. Note that Elliptic Curve signatures and key establishment schemes have been moved to ISO/IEC 14888 and ISO/IEC 11770 respectively together with other discrete log based mechanisms. Test vectors (examples) use P-256 are included for each parts for those mechanisms.*
- **SHA256:**
  - NIST-FIPS-180-4 (available at:  
<http://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.180-4.pdf>.)
  - ISO/IEC 10118-3 Hash-functions -- Part 3: Dedicated hash-functions (Clause 10)
- **OID** - ITU-T X.402 (available at: <https://www.itu.int/rec/T-REC-X.402-199906-I/en>.)
- **SP800-90A:**
  - NIST-SP-800-90A (available at:  
<http://csrc.nist.gov/publications/nistpubs/800-90A/SP800-90A.pdf>.)
    - *Note: NIST-SP-800-90A was withdrawn June 2015 and replaced by NIST-SP-800-90A Revision 1*  
<http://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-90Ar1.pdf>
- **SP800-90B** – NIST-SP-800-90B (available at:  
[http://csrc.nist.gov/publications/drafts/800-90/sp800-90b\\_second\\_draft.pdf](http://csrc.nist.gov/publications/drafts/800-90/sp800-90b_second_draft.pdf).)<sup>1</sup>

<sup>1</sup> Note that this document is still in DRAFT phase.

<sup>2</sup> Unless specified otherwise, all standards specified, including those from ISO, ITU, and NIST, refer to the version or edition which is more recent, as of 1 January 2016.

## 1.4 Terms and Abbreviations

This section defines the terms and abbreviations used throughout this document.

**Table 1-1: Terms and Abbreviations**

| Term/Abbreviation                    | Definition                                                                                                                                                                                      |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACD                                  | Additional Certificate Data for a Product.                                                                                                                                                      |
| Authentication                       | The process of determining whether an Entity is in fact who or what it claims to be.                                                                                                            |
| Authentication Initiator             | Refers to a Product that initiates Authentication.                                                                                                                                              |
| Authentication Responder             | Refers to a Product with whom an Authentication Initiator is attempting to authenticate.                                                                                                        |
| Certificate (Cert)                   | A digital form of identification that provides information about an Entity and certifies ownership of a particular public key.                                                                  |
| Certificate Authority (CA)           | An Entity that issues Certificates.                                                                                                                                                             |
| Certificate Chain                    | A series of two or more Certificates where each Certificate is signed by the preceding Certificate in the chain.                                                                                |
| Entity                               | Refers to a Product or an organization, vendor, or manufacturer associated with such Products.                                                                                                  |
| Evaluation Assurance Level (EAL)     | The numerical rating describing the depth and rigor of a security evaluation.                                                                                                                   |
| Intermediate Certificate             | A Certificate that is neither Root nor Leaf.                                                                                                                                                    |
| Leaf Certificate                     | The last Certificate in a Certificate Chain.                                                                                                                                                    |
| LSB                                  | Least Significant Byte                                                                                                                                                                          |
| MSB                                  | Most Significant Byte                                                                                                                                                                           |
| Nonce                                | A number used only once in any given key context. Can be interpreted as N-Once.                                                                                                                 |
| OID                                  | Object Identifier. See <b>OID</b> for more details.                                                                                                                                             |
| PD                                   | USB Power Delivery                                                                                                                                                                              |
| PD Product                           | Source, Sink, or Cable as defined in <b>USBPD</b>                                                                                                                                               |
| PDUSB Product                        | A USB Host or USB Device that is capable of both PD and USB communication.                                                                                                                      |
| Policy                               | Policy defines the behavior of Products. It defines the capabilities a Product advertises, its Authentication requirements, and resource availability with respect to unauthenticated Products. |
| Product                              | Refers to a PD Product, USB Product, and/or PDUSB Product.                                                                                                                                      |
| Pseudorandom Number Generator (PRNG) | A deterministic algorithm for generating a number or sequence of numbers that are computationally indistinguishable from truly random. See <b>SP800-90A</b> for more details.                   |
| Root Certificate                     | The first Certificate in a Certificate Chain. This certificate is self-signed.                                                                                                                  |
| TLV                                  | Type, Length, Value                                                                                                                                                                             |
| USB Device                           | A Device (including a Hub) as defined in <b>USB2.0</b> and <b>USB3.1</b> .                                                                                                                      |
| USB Host                             | A Host as defined in <b>USB2.0</b> and <b>USB3.1</b> .                                                                                                                                          |

Copyright © 2017 USB 3.0 Promoter Group. All rights reserved.

| Term/Abbreviation | Definition                                            |
|-------------------|-------------------------------------------------------|
| USB Hub           | A Hub as defined in <b>USB2.0</b> and <b>USB3.1</b> . |
| USB Product       | A USB Host or USB Device.                             |

## 1.5 Conventions

### 1.5.1 Precedence

If there is a conflict between text, figures, and tables, the precedence shall be tables, figures, and then text.

### 1.5.2 Keywords

The following keywords differentiate between the levels of requirements and options.

#### 1.5.2.1 Conditional Normative

**Conditional Normative** is a keyword used to indicate a feature that is mandatory when another related feature has been implemented. Designers are mandated to implement all such requirements, when the dependent features have been implemented, to ensure interoperability with other compliant Products.

#### 1.5.2.2 Deprecated

**Deprecated** is a keyword used to indicate a feature, supported in previous releases of the specification, which is no longer supported.

#### 1.5.2.3 Informative

**Informative** is a keyword that describes information with this specification that intends to discuss and clarify requirements and features as opposed to mandating them.

#### 1.5.2.4 May

**May** is a keyword that indicates a choice with no implied preference.

#### 1.5.2.5 N/A

**N/A** is a keyword that indicates that a field or value is not applicable and has no defined value and shall not be checked or used by the recipient.

#### 1.5.2.6 Normative

**Normative** is a keyword that describes features that are mandated by this specification.

#### 1.5.2.7 Optional/Optionally/Optional Normative

**Optional**, **Optionally**, and **Optional Normative** are equivalent keywords that describe features not mandated by this specification. However, if an **Optional** feature is implemented, the feature shall be implemented as defined by this specification.

#### 1.5.2.8 Reserved

**Reserved** is a keyword indicating reserved bits, bytes, words, fields, and code values that are set-aside for future standardization. Their use and interpretation may be specified by future extensions to this specification and, unless otherwise stated, shall not be utilized or adapted by vendor implementation. A **Reserved** bit, byte, word, or field shall be set to zero by the sender and shall be ignored by the receiver. **Reserved** field values shall not be sent by the sender and, if received, shall be ignored by the receiver.

#### 1.5.2.9 Shall/Normative

**Shall** and **Normative** are keywords indicating a mandatory requirement. Designers are mandated to implement all such requirements to ensure interoperability with other compliant Products.

#### 1.5.2.10 Should

**Should** is a keyword indicating flexibility of choice with a preferred alternative. Equivalent to the phrase “it is recommended that”.

### 1.5.3 Numbering

Numbers that are immediately followed by a lowercase “b” (e.g., 01b) are binary values. Numbers that are immediately followed by a lowercase “h” (e.g., 3Ah) are hexadecimal values. Numbers not immediately followed by either a “b”, or “h” are decimal values.

### 1.5.4 Byte Ordering

Unless otherwise specified, all multiple byte values in this specification are interpreted as and moved over the bus in little-endian order, i.e., LSB to MSB.

The order by which individual bits are moved over a bus is defined in **USBPD** for PD Products and **USB2.0** and **USB3.1** for USB Products.

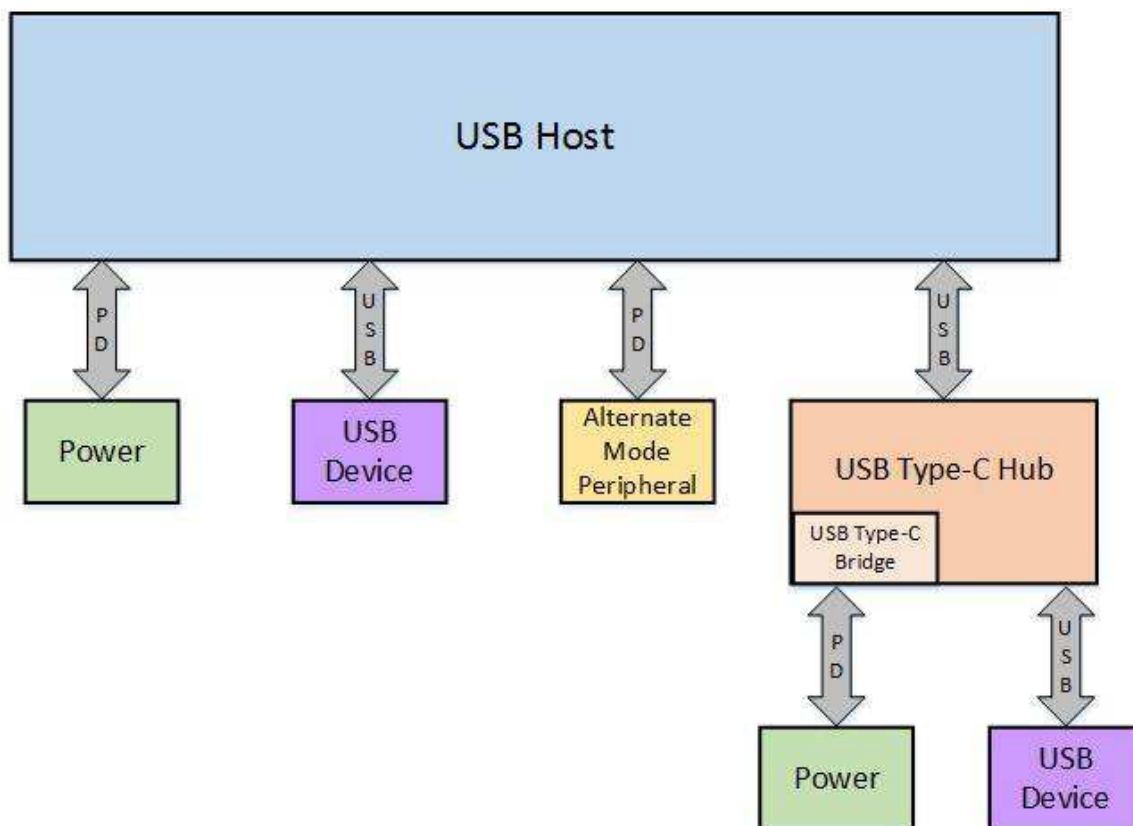
## 2 Overview

This section contains no Normative requirements.

### 2.1 Topology

Figure 2-1 represents a sample topology of USB Products and PD Products supporting USB Type-C Authentication. It is not meant to be comprehensive or represent the full range of topology configurations.

Figure 2-1 Sample Topology



## 2.2 Cryptographic Methods

This specification targets a 128-bit security level for all cryptographic methods. The cryptographic methods used by this specification are shown in Table 2-1.

Table 2-1: Summary of Cryptographic Methods

| Method                                                                                      | Use                                                         |
|---------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>X509v3</b> , <b>DER</b> encoding                                                         | Certificate format                                          |
| <b>ECDSA</b> using the <b>NIST P256</b> , <b>secp256r1</b> curve, uncompressed point format | Digital signing of Certificates and Authentication Messages |
| <b>SHA256</b>                                                                               | Hash algorithm                                              |

### 2.2.1 Random Numbers

The generation of cryptographic keys and the cryptographic protocol exchanges rely on cryptographic quality random numbers. Random numbers are defined as numbers that are distinguishably random by no algorithm with an algorithmic complexity of less than  $O(2^{128})$ .

The output of a NIST **SP800-90A** compliant PRNG seeded with a 256-bit full **SP800-90B** entropy value is sufficient to meet this standard.

## 2.3 Security Overview

This specification defines a Certificate-based method for Authentication that allows a Product to authenticate another attached Product and, by Policy, choose how to interact with that Product. For example, a PD Sink may choose not to use the full advertised capabilities of an unauthenticated PD Source. Authentication can be initiated by either a PD Sink, PD Source, or USB Host.

### 2.3.1 Periodic Re-Authentication

Products can optionally perform periodic re-Authentications. Re-Authentication is used to verify that an authenticated Product has not been replaced by a different Product.

### 2.3.2 Secret Key Storage and Protection

One threat concerning USB Type-C Authentication is the extraction of secret keys from Products. In a worst case scenario, the extraction of even one secret key could allow an attacker to clone Products in unlimited volume. This or similar scenarios would degrade trust in the whole USB Type-C Authentication ecosystem.

Therefore, it is recommended that vendors of Products take appropriate measures to protect the execution of the USB Type-C Authentication protocol and all private keys. Products should provide protected tamper-resistant operation and storage for the private keys to prevent them from being read (all or in part), copied or otherwise disclosed. This includes protection against side-channel and fault injection attacks, including software exploits and physical attacks such as leakage, probing, glitching, reverse engineering, and statistical analysis methods. Examples of such attacks include Simple and High-Order Differential Power, Electromagnetic, and Fault Analysis attacks. Other examples of attack vectors are listed in Appendix B.

### 2.3.3 Security Evaluation Criteria

The need for proven and measurable security evaluation results has led to worldwide established certification and evaluation schemes. One of the biggest and most widely applicable security evaluation schemes is **Common Criteria**, which provides global infrastructure for common recognition Certificates. This infrastructure includes government-driven supervision of certification authorities, accredited and capability balanced evaluation laboratories, and globally harmonized and internationally present mutual recognition contracts. In addition, **Common Criteria** implements a number of different evaluations from which a vendor is free to choose the appropriate level.

It is recommended that the vendor of a Product choose the market-driven individually required level of assurance (EAL), then conduct the independent evaluation process at an accredited evaluation laboratory. After evaluation, the result is published by the certification body in a Certificate which is automatically accepted by a high number of countries.

## 2.4 Impact to Existing Ecosystem

The impact to existing Products depends largely on individual Policy decisions regarding legacy products (i.e. products that predate this specification). For example, a USB Host with a Policy that allows full functioning of legacy devices will have a minimal impact on the current ecosystem while a USB Host with a Policy that limits or refuses to use functionalities exposed by a legacy product will have a more significant impact.



#### 2.4.1 Proxy Capabilities (PD traversing the Hub topology)

PD is a port-to-port communications protocol. Thus, in order to authenticate PD Products that are connected downstream of a USB Hub, it is necessary to implement a USB Type-C Bridge function in both the USB Hub and corresponding driver in the USB Host. The USB Type-C Bridge is used to translate PD requests to the USB data domain. See **USBTYPESPEC BRIDGE** for more information.

### 3 Authentication Architecture

#### 3.1 Certificates

##### 3.1.1 Format

All Certificates shall use the **X509v3 ASN.1** structure. All Certificates shall use binary **DER** encoding for **ASN.1**. All Certificates shall use the cryptographic methods listed in Table 2-1. The further description of the Certificate format assumes that the reader is familiar with **X509v3** Certificate terminology.

Leaf certificates shall not exceed *MaxLeafCertSize* in length. Intermediate Certificates shall not exceed *MaxIntermediateCertSize* in length.

Certificates and the fields, attributes, and extensions defined therein are Big Endian.

##### 3.1.2 Textual Format

All textual **ASN.1** objects contained within Certificates, including DirectoryString, GeneralName, and DisplayText, shall be specified as either a UTF8String, PrintableString, or IA5String. The length of any textual object shall not exceed 64 bytes excluding the **DER** type and **DER** length encoding.

##### 3.1.3 Attributes and Extensions

Where applicable, the Object Identifier (**OID**) is provided.

##### 3.1.3.1 Distinguished Name

The *distinguished name* consists of a number of attributes, which uniquely identify the Entity holding a corresponding private key. A Certificate Authority shall not issue Certificates with the same *distinguished name* to different Entities. *Distinguished name* uniqueness can be accomplished by including an attribute with unique values such as the binary X500UniqueIdentifier or textual *serial number* (See Section 3.1.3.1.3).

##### 3.1.3.1.1 Common Name (OID 2.5.4.3)

This attribute shall appear in every Certificate and shall contain a string matching one of the following three patterns:

- “USB:”
- “USB:<vid>”
- “USB:<vid>:<pid>”

Where <vid> represents a 4-character lowercase hexadecimal string encoding the 16-bit values corresponding to the VID of the Certificate *subject* and <pid> represents a 4-character lowercase hexadecimal string encoding the 16-bit values corresponding to the PID of the Certificate *subject*. When present, <vid> and <pid> shall be left zero padded and big endian. Uppercase letters shall not be used in the hex encoding of a VID or PID.

The *common name* attribute in the Leaf Certificate of a Certificate Chain shall contain both a VID and a PID. VID and PID are optional in the *common name* attribute of a non-Leaf Certificate. However, if a VID value appears in a Certificate in the Chain, then the same VID value shall be used in all subsequent Certificates. If a PID value appears in a Certificate in the Chain, then the same PID value shall be used in all subsequent Certificates.

#### 3.1.3.1.2 Organization Name (OID 2.5.4.10)

This attribute shall be present in a Root Certificate and may be present in other Certificates. When present, the *organization name* attribute shall contain the human-readable name of the organization that owns the private key that corresponds to the Certificate.

#### 3.1.3.1.3 Serial Number (OID 2.5.4.5)

This attribute is optional. If present, it shall only be present in a Leaf certificate. Different PD Products, even from the same production line, have different keys and therefore need distinct *distinguished names*. The *common name* attribute and *organization name* attribute do not provide such uniqueness. There are several attributes inside the *distinguished name* that can be used to make it unique. The recommended method is to use the *serial number* attribute to hold a lowercase hex-encoded value of the binary data (e.g. wafer number, lot number, production lot, etc.) necessary for uniqueness.

#### 3.1.3.2 Basic Constraints (OID 2.5.29.19)

This extension shall be present and marked as critical. The *cA* component shall be false in a Leaf Certificate. The *cA* component shall be true for a non-Leaf Certificate. Other components, including *pathLenConstraint*, shall not be included.

#### 3.1.3.3 Key Usage (OID 2.5.29.15)

This extension shall be present. Leaf Certificates shall have the *digitalSignature* bit set, and all other bits cleared. Non-Leaf Certificates shall have the *keyCertSign* bit set, may optionally have the *cRLSign* bit set, and shall have all other bits cleared.

#### 3.1.3.4 Extended Key Usage (OID 2.5.29.37)

This extension shall be present and marked as critical. It shall contain the USB-IF issued OID 2.23.145.1.1 for the “USB-Auth” extended key usage, and may contain other OIDs.

#### 3.1.3.5 Validity

The *notBefore* and *notAfter* fields indicate the time interval during which information regarding a Certificate’s validity is maintained. For Products, the validity times should be ignored.

Certificate *notBefore* and *notAfter* validity times shall be specified using either ASN.1 GeneralizedTime for any year, or ASN.1 UTCTime for years prior to 2050.

It is recommended that the *notBefore* field be “19700101000000Z” (for 00:00 on 01-Jan-1970 UTC, which is POSIX epoch time). It is recommended that the *notAfter* field be “99991231235959Z” (for 23:59:59 on 31-Dec-9999 UTC, which is used for an unknown expiration time as defined in IETF-RFC-5280, Section 4.1.2.5). Use of the recommended *notBefore* and *notAfter* values will maximize compatibility with certificate processing stacks.

### 3.1.3.6 USB-IF ACD (OID 2.23.145.1.2)

The USB-IF ACD extension is a custom Certificate extension for use with products compliant to this specification. It contains a binary object in the ACD format described in Appendix A.1. The binary object is encoded as an **ASN.1 DER** OCTET STRING with a maximum size of *MaxACDSIZE* bytes. The fields for a PD Product ACD are defined in Appendix A.2. The fields for a USB Product ACD are defined in Appendix A.3.

Leaf Certificates shall contain this extension. Non-Leaf Certificates shall not use this extension.

### 3.1.3.7 Additional Attributes and Extensions

Additional Certificate attributes and extensions defined in **X509v3** are allowed provided that the maximum Certificate size does not exceed *MaxLeafCertSize* for a Leaf Certificate or *MaxIntermediateCertSize* for a non-Leaf Certificate.

## 3.2 Certificate Chains

Certificates are grouped into Certificate Chains. A Certificate Chain is the binary (byte) concatenation of the fields shown in Table 3-1. A Certificate Chain shall not exceed *MaxCertChainSize* bytes.

**Table 3-1: Certificate Chain Format**

| Offset | Field               | Size        | Description                                                                                                                                                                                                                                                                       |
|--------|---------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | <i>Length</i>       | 2           | Total length of Certificate Chain in bytes including all fields in this table.<br>This field is little endian.                                                                                                                                                                    |
| 2      | <i>Reserved</i>     | 2           | Set to zero                                                                                                                                                                                                                                                                       |
| 4      | <i>RootHash</i>     | 32          | 32-byte <b>SHA256</b> hash of the Root Certificate.<br>This field is big endian.                                                                                                                                                                                                  |
| 36     | <i>Certificates</i> | Length - 36 | One or more <b>ASN.1 DER</b> -encoded <b>X509v3</b> Certificates where the first Certificate is signed by the Root Certificate and each subsequent Certificate is signed by the preceding Certificate. The last Certificate is the Leaf Certificate.<br>This field is big endian. |

Certificate Chains reside in positions called slots. Each slot shall either be empty or contain one complete certificate chain. A Product shall not contain more than 8 slots. Slots 0 through 3 shall only be used for Certificate Chains rooted with a USB-IF Root Certificate and shall not contain any other Certificate Chains. Slots 4 through 7 may be used for any additional Certificate Chains. The **ASN.1 DER** encoding of each individual certificate can be analyzed to determine its length.

### 3.2.1 Provisioning

Provisioning is the process by which a Product acquires one or more Certificate Chains. The procedure for provisioning a Product is outside the scope of this specification.

### 3.3 Private Keys

Each Certificate Chain in a Product corresponds to a private key whose corresponding public key is certified in the Leaf Certificate of that slot. The Product must have access to that private key. Private keys shall be generated, provisioned, and stored in a manner that adequately protects the confidentiality of the key.

The following rules govern private key uniqueness:

- A private key used by one Product shall not be used by any other Products.
- A private key used by a Product in one slot shall not be used in any other slots.

## 4 Authentication Protocol

There are three operations an Authentication Initiator can perform:

- Query an Authentication Responder for Certificate Chain digests
- Read a Certificate Chain from an Authentication Responder
- Challenge an Authentication Responder in order to verify its authenticity

An Authentication Initiator may initiate as many or as few of these operation as are needed to achieve the desired Authentication latency. In addition, an Authentication Initiator may initiate the operations in any order. For example, an Authentication Initiator that only uses slot 0 for Authentication, may first challenge a PD Product, and then initiate a Certificate Chain read if the target Certificate Chain is not already cached.

A Product shall not act as an Authentication Responder unless it contains a Certificate Chain in slot 0.

### 4.1 Digest Query

To query an Authentication Responder for Certificate Chain digests, an Authentication Initiator sends a GET\_DIGESTS Request as defined in Section 5.2.1. If an error condition is encountered, the Authentication Responder shall respond with the appropriate ERROR Response as defined in Section 4.4. Otherwise, the Authentication Responder shall respond with a DIGESTS Response as defined in Section 5.3.1. After receiving a DIGESTS Response, an Authentication Initiator can check to see if it has any of the Authentication Responder's Certificate Chains cached. This allows the Authentication Initiator to potentially skip reading a Certificate Chain and thus save time.

### 4.2 Certificate Chain Read

To read a Certificate Chain, or portion thereof, an Authentication Initiator sends a GET\_CERTIFICATE Request as defined in Section 5.2.2.

If an Authentication Responder receives a GET\_CERTIFICATE request that targets an offset that is outside the Certificate Chain (i.e.  $\text{offset} > \text{length}$ ) or attempts to read beyond the length of the target Certificate Chain (i.e.  $(\text{offset} + \text{length}) > \text{Certificate Chain length}$ ), then the Authentication Responder shall return an ERROR Authentication Response with *Param1* set to INVALID\_REQUEST and *Param2* set to 00h.

If an error condition is encountered, the Authentication Responder shall respond with the appropriate ERROR Response as defined in Section 4.4. Otherwise, the Authentication Responder shall respond with a CERTIFICATE Response as described in Section 5.3.2.

### 4.3 Authentication Challenge

To challenge an Authentication Responder, an Authentication Initiator sends a CHALLENGE Request as defined in Section 5.2.3. If an error condition is encountered, the Authentication Responder shall respond with the appropriate ERROR Response as defined in Section 4.4. Otherwise, the Authentication Responder shall respond with a CHALLENGE\_AUTH Response as described in Section 5.3.3.

### 4.4 Errors and Alerts

#### 4.4.1 Invalid Request

If an Authentication Responder receives an Authentication Request with one or more invalid fields, it shall respond to that Authentication Request with an ERROR Response that has *Param1* set to INVALID\_REQUEST and *Param2* set to 00h.

#### 4.4.2 Unsupported Protocol Version

If an Authentication Responder receives an Authentication Request that contains an unsupported Security Protocol Version in the *ProtocolVersion* field, it shall respond to that Authentication Request with an ERROR Response that has *ProtocolVersion* set to the minimum Security Protocol Version it supports, *Param1* set to UNSUPPORTED\_PROTOCOL, and *Param2* set to the maximum Security Protocol Version it supports.

#### 4.4.3 Busy

If an Authentication Responder receives an Authentication Request but is unable to meet either the timing requirements listed in Section 6.3 (for PD Products) or Section 7.4 (for USB Products), it shall respond to that Authentication Request with an ERROR Response that has *Param1* set to BUSY and *Param2* set to 00h.

#### 4.4.4 Unspecified

If an Authentication Responder receives an Authentication Request then encounters an error that is not covered by the conditions above, it shall respond to that Authentication Request with an ERROR Response that has *Param1* set to UNSPECIFIED and *Param2* set to 00h.

## 5 Authentication Messages

Authentication Messages are used to convey information related to Authentication. An Authentication Message consists of a Message Header followed by a variable length (including zero) payload. The format for a Message Header is defined in Section 5.1.

There are two types of Authentication Messages: Authentication Requests and Authentication Responses. Authentication Requests are defined in Section 5.2. Authentication Responses are defined in Section 5.3.

Section 6 describes how Authentication Messages map onto PD messaging. Section 7 describes how Authentication Messages map onto USB transfers.

## 5.1 Header

All Authentication Messages shall start with the 4-byte header defined in Table 5-1.

**Table 5-1: Authentication Message Header**

| Offset | Field                  | Size | Reference     |
|--------|------------------------|------|---------------|
| 0      | <i>ProtocolVersion</i> | 1    | Section 5.1.1 |
| 1      | <i>MessageType</i>     | 1    | Section 5.1.2 |
| 2      | <i>Param1</i>          | 1    | Section 5.1.3 |
| 3      | <i>Param2</i>          | 1    | Section 5.1.4 |

### 5.1.1 USB Type-C Authentication Protocol Version

This field identifies which version of the USB Type-C Authentication Specification is being used. Table 5-2 shows the valid values for this field. A Product shall not use a USB Type-C Authentication Protocol Version value corresponding to a specification revision that it does not support.

**Table 5-2: USB Type-C Authentication Protocol Version**

| Name     | Value   | Meaning                                        |
|----------|---------|------------------------------------------------|
| Reserved | 00h     | Reserved                                       |
| V1.0     | 01h     | USB Type-C Authentication Protocol Version 1.0 |
| Reserved | 02h-ffh | Reserved                                       |

It is intended in the future that Products support a contiguous range of USB Type-C Authentication Protocol Versions.

### 5.1.2 Message Type

This field identifies Authentication Message type and shall contain one of the Authentication Message Types listed in Table 5-3 or Table 5-9.

### 5.1.3 Param1

This field is used to pass a first 1-byte parameter. The contents of the parameter vary and are defined by Authentication Message type.

### 5.1.4 Param2

This field is used to pass a second 1-byte parameter. The contents of the parameter vary and are defined by Authentication Message type.

## 5.2 Authentication Requests

Authentication Requests are used by an Authentication Initiator to send a command to an Authentication Responder and/or retrieve data. Authentication Request types are listed in Table 5-3.

An Authentication Initiator shall not send another Authentication Request until it has either received a response for or timed out the previously sent Authentication Request.

**Table 5-3: Authentication Request Types**

| Value     | Description                                     |
|-----------|-------------------------------------------------|
| 00h – 7Fh | Shall only be used for Authentication Responses |
| 80h       | Reserved                                        |
| 81h       | GET_DIGESTS                                     |
| 82h       | GET_CERTIFICATE                                 |
| 83h       | CHALLENGE                                       |
| 84h - FFh | Reserved                                        |

### 5.2.1 GET\_DIGESTS

This Request is used to retrieve Certificate Chain digests. The header for a GET\_DIGESTS Request is defined in Table 5-4. A GET\_DIGESTS Request has no payload.

**Table 5-4: GET\_DIGESTS Request Header**

| Offset | Field                  | Size | Value       |
|--------|------------------------|------|-------------|
| 0      | <i>ProtocolVersion</i> | 1    | V1.0        |
| 1      | <i>MessageType</i>     | 1    | GET_DIGESTS |
| 2      | <i>Param1</i>          | 1    | Reserved    |
| 3      | <i>Param2</i>          | 1    | Reserved    |

### 5.2.2 GET\_CERTIFICATE

This Request is used to read a segment of a target Certificate Chain. The header for a GET\_CERTIFICATE Request is defined in Table 5-5. The payload for a GET\_CERTIFICATE Request is defined in Table 5-6.

**Table 5-5: GET\_CERTIFICATE Request Header**

| Offset | Field                  | Size | Value                                                                                                                 |
|--------|------------------------|------|-----------------------------------------------------------------------------------------------------------------------|
| 0      | <i>ProtocolVersion</i> | 1    | V1.0                                                                                                                  |
| 1      | <i>MessageType</i>     | 1    | GET_CERTIFICATE                                                                                                       |
| 2      | <i>Param1</i>          | 1    | Slot number of the target Certificate Chain to read from. The value in this field shall be between 0 and 7 inclusive. |
| 3      | <i>Param2</i>          | 1    | Reserved                                                                                                              |

**Table 5-6: GET\_CERTIFICATE Request Payload**

| Offset | Field         | Size | Value                                                                                                                     |
|--------|---------------|------|---------------------------------------------------------------------------------------------------------------------------|
| 4      | <i>Offset</i> | 2    | Offset in bytes from the start of the Certificate Chain to where the read request begins.<br>This field is little endian. |
| 6      | <i>Length</i> | 2    | Length in bytes of the read request.<br>This field is little endian.                                                      |

### 5.2.3 CHALLENGE

This Request is used to initiate Authentication of a Product. The header for a CHALLENGE Request is defined in Table 5-7. The payload for a CHALLENGE Request is defined in Table 5-8.

**Table 5-7: CHALLENGE Request Header**

| Offset | Field                  | Size | Value                                                                                 |
|--------|------------------------|------|---------------------------------------------------------------------------------------|
| 0      | <i>ProtocolVersion</i> | 1    | V1.0                                                                                  |
| 1      | <i>MessageType</i>     | 1    | CHALLENGE                                                                             |
| 2      | <i>Param1</i>          | 1    | Slot number of the recipient's Certificate Chain that will be used for Authentication |
| 3      | <i>Param2</i>          | 1    | Reserved                                                                              |

**Table 5-8: CHALLENGE Request Payload**

| Offset | Field        | Size | Description                                                                                  |
|--------|--------------|------|----------------------------------------------------------------------------------------------|
| 4      | <i>Nonce</i> | 32   | Random 32-byte nonce chosen by the Authentication Initiator.<br>This field is little endian. |

### 5.3 Authentication Responses

Authentication Responses are used by an Authentication Responder to respond to an Authentication Request. Authentication Response types are listed in Table 5-9.

**Table 5-9: Authentication Response Types**

| Value     | Description                                    |
|-----------|------------------------------------------------|
| 00h       | Reserved                                       |
| 01h       | DIGESTS                                        |
| 02h       | CERTIFICATE                                    |
| 03h       | CHALLENGE_AUTH                                 |
| 04h-7Eh   | Reserved                                       |
| 7Fh       | ERROR                                          |
| 80h – FFh | Shall only be used for Authentication Requests |



### 5.3.1 DIGESTS

This Response is used by a Product to send Certificate Chain digests and report which slots contain valid Certificate Chain digests. The header for a DIGESTS Response is defined in Table 5-10. The Payload for a DIGESTS Response is defined in Table 5-11.

**Table 5-10: DIGESTS Response Header**

| Offset | Field                  | Size | Value                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------|------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | <i>ProtocolVersion</i> | 1    | V1.0                                                                                                                                                                                                                                                                                                                                                                                                      |
| 1      | <i>MessageType</i>     | 1    | DIGESTS                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2      | <i>Param1</i>          | 1    | Capabilities Field; shall be set to 01h for this specification. All other values reserved.                                                                                                                                                                                                                                                                                                                |
| 3      | <i>Param2</i>          | 1    | Slot mask. The bit in position K of this byte shall be set to 1b if and only if slot number K contains a Certificate Chain for the protocol version in the <i>ProtocolVersion</i> field. (Bit 0 is the least significant bit of the byte.)<br><br>The number of digests returned shall be equal to the number of bits set in this byte. The digests shall be returned in order of increasing slot number. |

**Table 5-11: DIGESTS Response Payload**

| Offset                | Field              | Size | Value                                                                                                 |
|-----------------------|--------------------|------|-------------------------------------------------------------------------------------------------------|
| 4                     | <i>Digest[0]</i>   | 32   | 32-byte SHA-256 digest of the first Certificate Chain.<br>This field is big endian.                   |
| ...                   | ...                | ...  | ...                                                                                                   |
| 4 +<br>(32 * (n - 1)) | <i>Digest[n-1]</i> | 32   | 32-byte SHA-256 digest of the last (n <sup>th</sup> ) Certificate Chain.<br>This field is big endian. |

### 5.3.2 CERTIFICATE

This Response is used by a Product to send the requested segment of a Certificate Chain. The header for a CERTIFICATE Response is defined in Table 5-12. The payload for a CERTIFICATE Response is defined in Table 5-13.

**Table 5-12: CERTIFICATE Response Header**

| Offset | Field                  | Size | Value                                         |
|--------|------------------------|------|-----------------------------------------------|
| 0      | <i>ProtocolVersion</i> | 1    | V1.0                                          |
| 1      | <i>MessageType</i>     | 1    | CERTIFICATE                                   |
| 2      | <i>Param1</i>          | 1    | Slot number of the Certificate Chain returned |
| 3      | <i>Param2</i>          | 1    | Reserved                                      |

**Table 5-13: CERTIFICATE Response Payload**

| Offset | Field            | Size          | Value | Description                                                                                                            |
|--------|------------------|---------------|-------|------------------------------------------------------------------------------------------------------------------------|
| 4      | <i>CertChain</i> | <i>Length</i> | Data  | Requested contents of target Certificate Chain.<br><br>The endianness for a Certificate Chain is defined in Table 3-1. |

**5.3.3 CHALLENGE\_AUTH**

This Response is used by a Product to respond to a CHALLENGE Request. The header for a CHALLENGE\_AUTH Response is defined in Table 5-14. The payload for a CHALLENGE\_AUTH Response is defined in Table 5-15.

**Table 5-14: CHALLENGE\_AUTH Response Header**

| Offset | Field                  | Size | Value                                                                                                                                                                                                                                      |
|--------|------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | <i>ProtocolVersion</i> | 1    | V1.0                                                                                                                                                                                                                                       |
| 1      | <i>MessageType</i>     | 1    | CHALLENGE_AUTH                                                                                                                                                                                                                             |
| 2      | <i>Param1</i>          | 1    | Shall contain the Slot number in the <i>Param1</i> field of the corresponding CHALLENGE Request                                                                                                                                            |
| 3      | <i>Param2</i>          | 1    | Slot mask. The bit in position K of this byte shall be set to 1b if and only if slot number K contains a Certificate Chain for the protocol version in the <i>ProtocolVersion</i> field. (Bit 0 is the least significant bit of the byte.) |

**Table 5-15: CHALLENGE\_AUTH Response Payload**

| Offset | Field                     | Size | Value                                                                                                                                                   |
|--------|---------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4      | <i>MinProtocolVersion</i> | 1    | Minimum protocol version supported by this Device                                                                                                       |
| 5      | <i>MaxProtocolVersion</i> | 1    | Maximum protocol version supported by this Device                                                                                                       |
| 6      | <i>Capabilities</i>       | 1    | Set to 01h for this specification. All other values reserved                                                                                            |
| 7      | <i>Reserved</i>           | 1    | Reserved                                                                                                                                                |
| 8      | <i>CertChainHash</i>      | 32   | 32-byte <b>SHA256</b> hash of the Certificate Chain used for Authentication.<br>This field is big endian.                                               |
| 40     | <i>Salt</i>               | 32   | 32-byte value chosen by the Authentication Responder.<br>This field is little endian.<br><i>Note: the Salt can be random, fixed, or any other value</i> |
| 72     | <i>Context Hash</i>       | 32   | See Section 6.4 for PD Products or Section 7.5 for USB Products.<br>This field is big endian.                                                           |
| 104    | <i>Signature</i>          | 64   | See Section 5.3.3.1.<br>This field is little endian.                                                                                                    |

#### 5.3.3.1 Signature

The *Signature* field in a CHALLENGE\_AUTH Response contains a 64-byte **ECDSA** digital signature on the message contents listed in Table 5-16. The **ECDSA** signature is generated using values (r,s) with little-endian encoding, where r starts at offset 0 and s starts at offset 32. Each value is 32 Bytes with zero right-padding if necessary.

**Table 5-16: Message Contents for ECDSA Digital Signature**

| Offset | Field          | Size | Value                                                                                     |
|--------|----------------|------|-------------------------------------------------------------------------------------------|
| 0      | <i>ReqMsg</i>  | 36   | Full contents (i.e. header and payload) of the corresponding CHALLENGE Request            |
| 36     | <i>RespMsg</i> | 104  | Contents of the CHALLENGE_AUTH Response being signed excluding the <i>Signature</i> field |

A message signer with a secure RNG can use non-deterministic **ECDSA**. A message signer without secure RNG capability can use deterministic **ECDSA**. In deterministic **ECDSA**, the random “k” value is derived from the hash of the message to be signed and a private key.

*Note: in both deterministic and non-deterministic ECDSA, generating the “k” value has pitfalls and mistakes can lead to a leak of the private key. See RFC 6979 (available at: <https://tools.ietf.org/html/rfc6979>) for details.*

#### 5.3.4 ERROR

This Response is used by a Product to transmit error information. The header for an ERROR Response is defined in Table 5-17. An ERROR Response has no payload.

**Table 5-17: ERROR Response Header**

| Offset | Field                  | Size | Description                 |
|--------|------------------------|------|-----------------------------|
| 0      | <i>ProtocolVersion</i> | 1    | V1.0 <sup>1</sup>           |
| 1      | <i>MessageType</i>     | 1    | ERROR                       |
| 2      | <i>Param1</i>          | 1    | Error Code. See Table 5-18. |
| 3      | <i>Param2</i>          | 1    | Error Data. See Table 5-18. |

**Table 5-18: ERROR Codes**

| Error Code           | Value       | Description                                                          | Error Data                                   |
|----------------------|-------------|----------------------------------------------------------------------|----------------------------------------------|
| Reserved             | 00h         | Reserved                                                             | Reserved                                     |
| INVALID_REQUEST      | 01h         | One or more Request fields are invalid                               | 00h                                          |
| UNSUPPORTED_PROTOCOL | 02h         | Requested Security Protocol Version is not supported                 | Maximum supported Security Protocol Version1 |
| BUSY                 | 03h         | Device cannot respond now, but will be able to respond in the future | 00h                                          |
| UNSPECIFIED          | 04h         | Unspecified error occurred                                           | 00h                                          |
| Reserved             | 05h-<br>EFh | Reserved                                                             | Reserved                                     |
| Vendor Defined       | F0h-<br>FFh | Vendor defined                                                       | Vendor defined                               |

<sup>1</sup>Note: Minimum supported Security Protocol Version is returned in the *ProtocolVersion* field in the message header.

## 6 Authentication of PD Products

**USBPD** describes the mechanism for sending Authentication Messages over PD. The mapping of a security transfer to PD messages depends on whether or not the transfer exceeds *MaxExtendedMsgLen*. Note that *MaxExtendedMsgLen* is defined in **USBPD**.

### 6.1 Transfers less than or equal to *MaxExtendedMsgLen*

Security transfers where the Authentication Request and corresponding Authentication Response are each less than or equal to *MaxExtendedMsgLen* require only a single PD Message exchange. To initiate a security transfer, an Authentication Initiator sends a Security\_Request PD Message carrying an Authentication Request in the PD Authentication Request Data Block (SRQDB). To complete a security transfer, an Authentication Responder sends a Security\_Response PD Message carrying an Authentication Response in the PD Authentication Response Data Block (SRPDB).

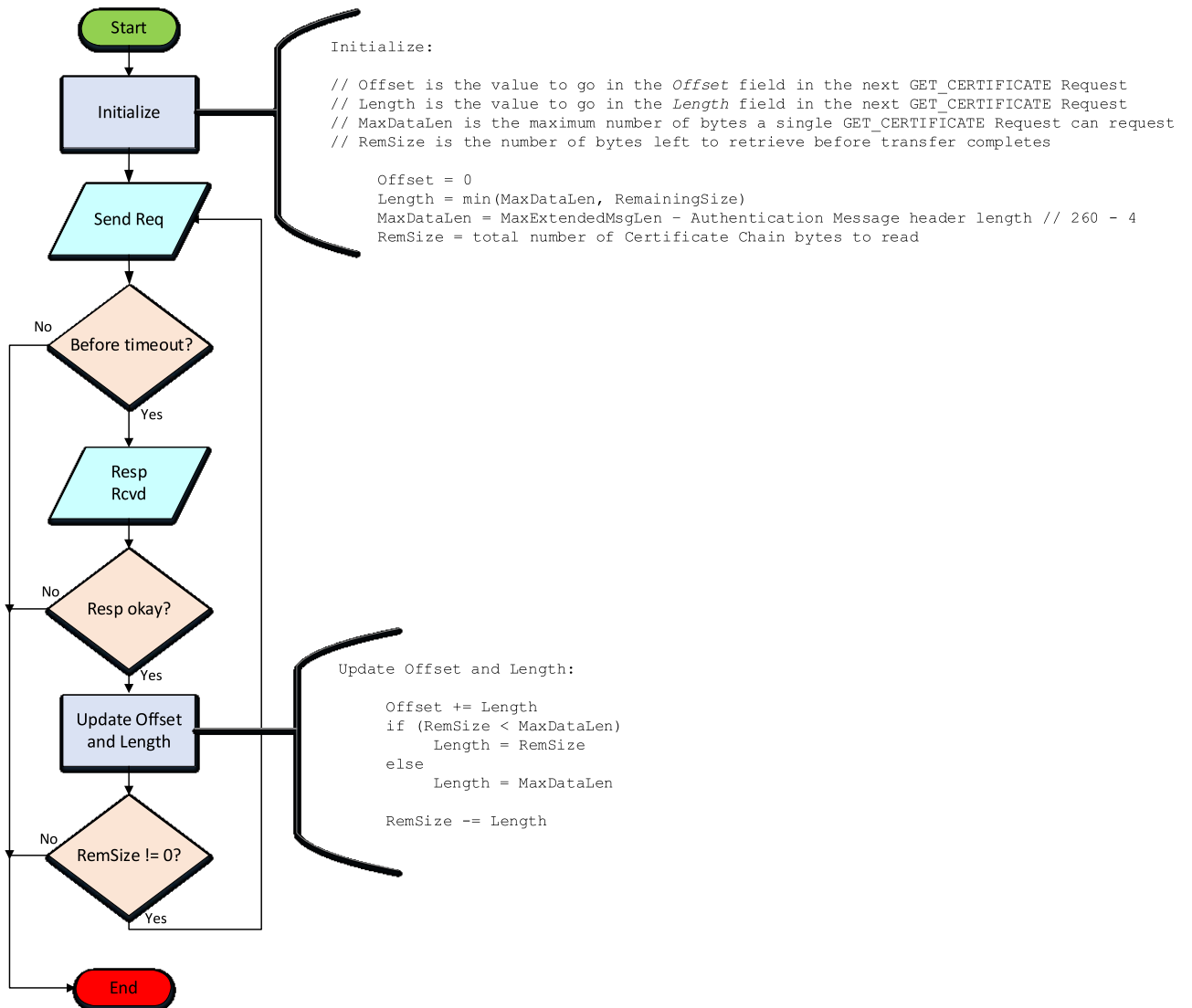
For example, to retrieve Certificate Chain digests, an Authentication Initiator sends a Security\_Request PD Message carrying a GET\_DIGESTS Authentication Message. The Authentication Responder then responds with a Security\_Response PD Message carrying a DIGESTS Authentication Message (or appropriate ERROR Authentication Message).

## 6.2 Transfers greater than *MaxExtendedMsgLen*

Security transfers where the Authentication Response is greater than *MaxExtendedMsgLen* require multiple PD message exchanges in order to fit into the PD Message framework.

An Authentication Initiator shall break up a security transfer into Authentication Messages that don't exceed *MaxExtendedMsgLen*. An Authentication Initiator does this by sending a series of Authentication Requests that target an incrementing segment of the security data to read in. Each Authentication Request is sent in the SRQDB of a Security\_Request PD Message. Each Authentication Response is sent as in the SRPDB of a Security\_Response PD Message. Figure 6-1 shows an example of how an Authentication Initiator can break up and complete a security transfer when more than *MaxExtendedMsgLen* number of bytes are read from the Certificate Chain of an Authentication Responder.

**Figure 6-1 Example Security Transfer Process for an Authentication Initiator**



*Note: timeout values are listed in Table 6-1.*

Figure 6-2 is an example of how an Authentication Responder can respond to a GET\_CERTIFICATE Request. The criteria the Authentication Responder uses to determine whether or not a given GET\_CERTIFICATE Request is valid is listed in the pseudocode at the right of the figure.

**Figure 6-2 Example Security Transfer Process for an Authentication Responder**

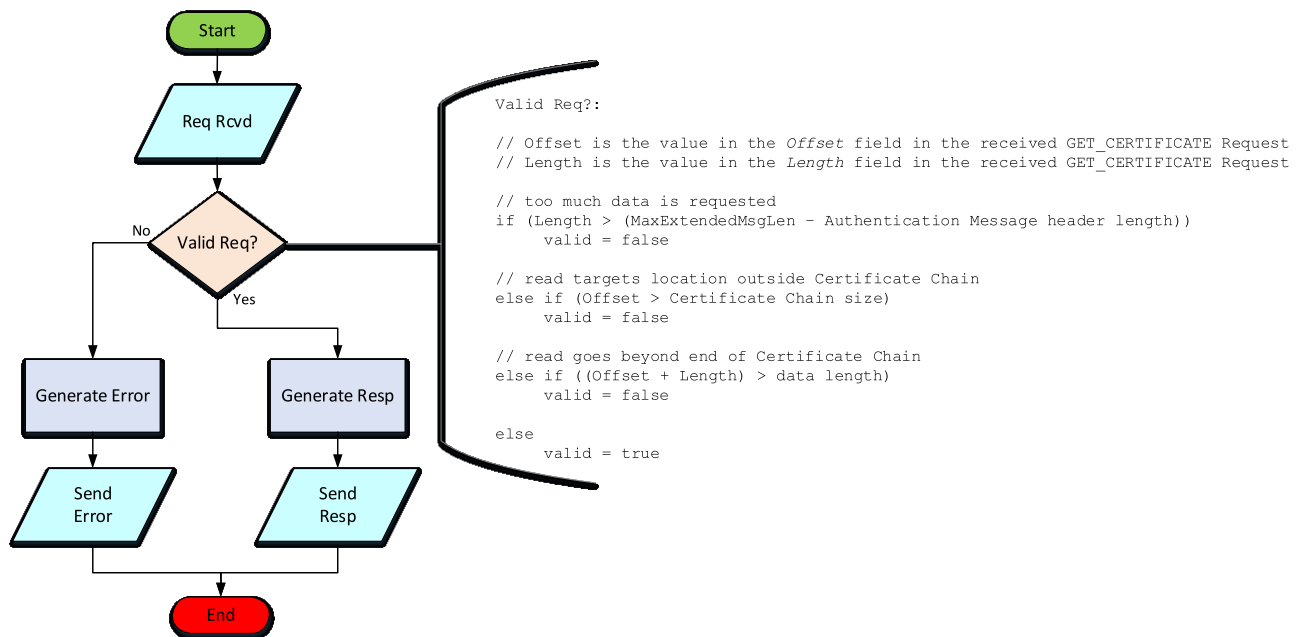
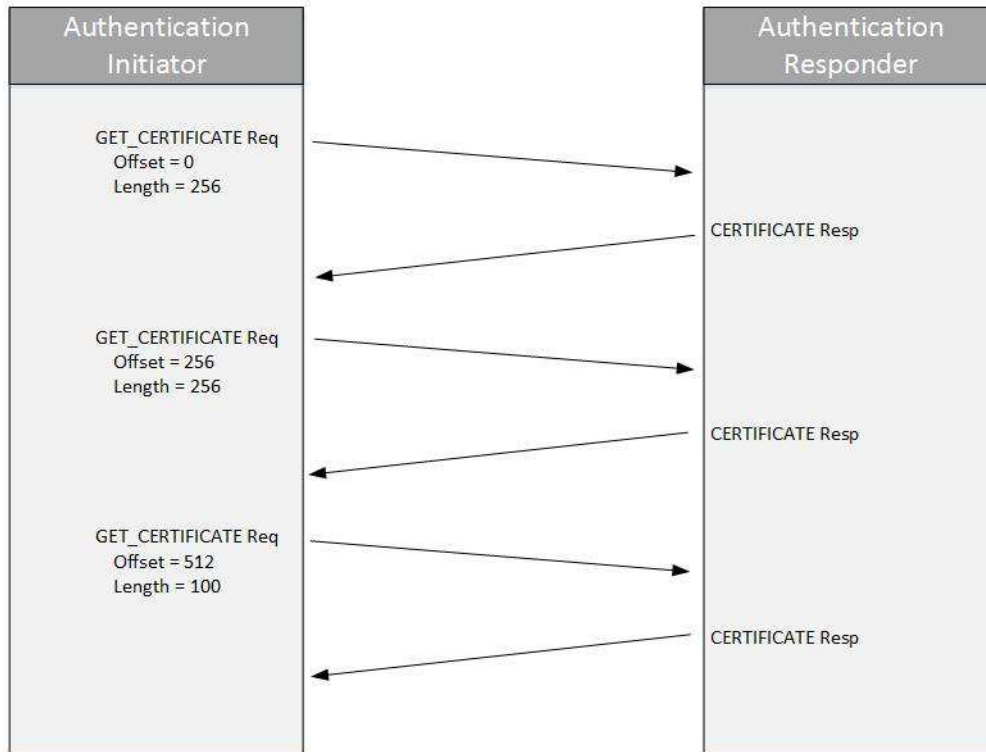


Figure 6-3 shows an example of a GET\_CERTIFICATE Request for a 612-Byte Certificate Chain using the processes in Figure 6-1 and Figure 6-2.

**Figure 6-3 Example 612-Byte Certificate Chain Read**



### 6.3 Timing Requirements for PD Security Extended Messages

#### 6.3.1 Authentication Initiator

Table 6-1 shows the timeout values that apply to an Authentication Initiator. The timeout values shall be measured from when the last bit of the **GoodCRC** Message **EOP**, corresponding to the Authentication Request Message, has been received by the PD Physical Layer until the last bit of the Authentication Response Message **EOP** has been received by the PD Physical Layer.

**Table 6-1: Timeout Values for a PD Authentication Initiator**

| Parameter                 | Timeout Value |         | Description                                           |
|---------------------------|---------------|---------|-------------------------------------------------------|
|                           | Unchunked     | Chunked |                                                       |
| <i>tDigestRcvd</i>        | 40 ms         | 200 ms  | Timeout for a GET_DIGESTS Authentication Request.     |
| <i>tCertRcvd</i>          | 40 ms         | 200 ms  | Timeout for a GET_CERTIFICATE Authentication Request. |
| <i>tChallengeAuthRcvd</i> | 1000 ms       | 1200 ms | Timeout for a CHALLENGE Authentication Request.       |



If an Authentication Initiator does not receive an Authentication Response within *tDigestRcvd* of sending a GET\_DIGESTS Authentication Request, it is considered an error. Subsequent handling of this error is outside the scope of this specification.

If an Authentication Initiator does not receive an Authentication Response within *tCertRcvd* of sending a GET\_CERTIFICATE Authentication Request to an Authentication Responder, it is considered an error. Subsequent handling of this error is outside the scope of this specification.

If an Authentication Initiator does not receive an Authentication Response within *tChallengeAuthRcvd* of sending a CHALLENGE Authentication Request, it is considered an error. Subsequent handling of this error is outside the scope of this specification.

### 6.3.2 Authentication Responder

Table 6-2 gives timing requirements that an Authentication Responder shall meet. The timing requirements shall be measured from when the last bit of the Authentication Request Message **EOP** has been received by the PD Physical Layer until the last bit of the **GoodCRC** Message **EOP** corresponding to the Authentication Response Message has been received by the PD Physical Layer.

**Table 6-2: Timing Requirements for PD Authentication Responder**

| Parameter                 | Timing Value |         | Description                                                                                                     |
|---------------------------|--------------|---------|-----------------------------------------------------------------------------------------------------------------|
|                           | Unchunked    | Chunked |                                                                                                                 |
| <i>tDigestSent</i>        | 20 ms        | 135 ms  | Maximum time between receiving a GET_DIGESTS Authentication Request and sending an Authentication Response.     |
| <i>tCertSent</i>          | 20 ms        | 135 ms  | Maximum time between receiving a GET_CERTIFICATE Authentication Request and sending an Authentication Response. |
| <i>tChallengeAuthSent</i> | 520 ms       | 635 ms  | Maximum time between receiving a CHALLENGE Authentication Request and sending an Authentication Response.       |

An Authentication Response shall be sent within *tDigestSent* of receiving a GET\_DIGESTS Authentication Request.

An Authentication Response shall be sent within *tCertSent* of receiving a GET\_CERTIFICATE Authentication Request.

An Authentication Response shall be sent within *tChallengeAuthSent* of receiving a CHALLENGE Authentication Request.

## 6.4 Context Hash

The *Context Hash* field in a CHALLENGE\_AUTH Authentication Response shall be zero for PD Sources, Sinks and Cable Plugs. This field shall also be zero for PD Alternate Mode devices.

## 7 Authentication of USB Products

A USB Host can use the architecture and methods defined in this specification to authenticate a USB Device. A USB Host that authenticates a USB Device takes the role of Authentication Initiator with the USB Device taking the role of Authentication Responder. A USB Device shall not act as an Authentication Initiator.

This section describes the additional framework needed for USB Device Authentication.

### 7.1 Descriptors

The descriptors in this section are used to describe the Authentication capabilities of a USB Device.

#### 7.1.1 Authentication Capability Descriptor

This descriptor is used to advertise the Authentication capabilities and features a USB Device supports. This descriptor shall be returned as part of the BOS Descriptor set for a USB Device that supports Authentication.

**Table 7-1: Authentication Capability Descriptor**

| Offset | Field                                                                                                       | Size | Value    | Description                                                                                                                                                                                                                                                                                                                                                                                                                 |     |             |   |                                                                                              |   |                                                                                                             |     |                                 |
|--------|-------------------------------------------------------------------------------------------------------------|------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------------|---|----------------------------------------------------------------------------------------------|---|-------------------------------------------------------------------------------------------------------------|-----|---------------------------------|
| 0      | <i>bLength</i>                                                                                              | 1    | Number   | Size of this Descriptor in Bytes                                                                                                                                                                                                                                                                                                                                                                                            |     |             |   |                                                                                              |   |                                                                                                             |     |                                 |
| 1      | <i>bDescriptorType</i>                                                                                      | 1    | Constant | Descriptor type: DEVICE CAPABILITY (16)                                                                                                                                                                                                                                                                                                                                                                                     |     |             |   |                                                                                              |   |                                                                                                             |     |                                 |
| 2      | <i>bDevCapabilityType</i>                                                                                   | 1    | Number   | AUTHENTICATION                                                                                                                                                                                                                                                                                                                                                                                                              |     |             |   |                                                                                              |   |                                                                                                             |     |                                 |
| 3      | <i>bmAttributes</i>                                                                                         | 1    | Bitmap   | Bitmap encoding of supported features. <table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>Shall be set to 1 to indicate that firmware can be updated. Otherwise, shall be set to zero.</td></tr><tr><td>1</td><td>Shall be set to 1 to indicate that Device changes interfaces when updated. Otherwise, shall be set to zero.</td></tr><tr><td>7:2</td><td>Reserved. Shall be set to zero.</td></tr></table> | Bit | Description | 0 | Shall be set to 1 to indicate that firmware can be updated. Otherwise, shall be set to zero. | 1 | Shall be set to 1 to indicate that Device changes interfaces when updated. Otherwise, shall be set to zero. | 7:2 | Reserved. Shall be set to zero. |
| Bit    | Description                                                                                                 |      |          |                                                                                                                                                                                                                                                                                                                                                                                                                             |     |             |   |                                                                                              |   |                                                                                                             |     |                                 |
| 0      | Shall be set to 1 to indicate that firmware can be updated. Otherwise, shall be set to zero.                |      |          |                                                                                                                                                                                                                                                                                                                                                                                                                             |     |             |   |                                                                                              |   |                                                                                                             |     |                                 |
| 1      | Shall be set to 1 to indicate that Device changes interfaces when updated. Otherwise, shall be set to zero. |      |          |                                                                                                                                                                                                                                                                                                                                                                                                                             |     |             |   |                                                                                              |   |                                                                                                             |     |                                 |
| 7:2    | Reserved. Shall be set to zero.                                                                             |      |          |                                                                                                                                                                                                                                                                                                                                                                                                                             |     |             |   |                                                                                              |   |                                                                                                             |     |                                 |
| 4      | <i>bcdProtocolVersion</i>                                                                                   | 1    | Number   | Shall be set to the USB Type-C Authentication Protocol Version                                                                                                                                                                                                                                                                                                                                                              |     |             |   |                                                                                              |   |                                                                                                             |     |                                 |
| 5      | <i>bcdCapability</i>                                                                                        | 1    | Number   | Shall be set to the same value as <i>Param1</i> in a DIGESTS Authentication Response                                                                                                                                                                                                                                                                                                                                        |     |             |   |                                                                                              |   |                                                                                                             |     |                                 |

**Table 7-2: Authentication Capability Descriptor Types**

| Authentication Capability Descriptor Type | Value |
|-------------------------------------------|-------|
| AUTHENTICATION                            | 0Eh   |

## 7.2 Mapping Authentication Messages to USB

Authentication Messages are transmitted over USB using the control requests defined in this section. Two additional USB Standard Request Codes used for USB Authentication are defined in Table 7-3.

**Table 7-3: Authentication Message *bRequest* Values**

| Authentication Message <i>bRequest</i> | Value |
|----------------------------------------|-------|
| AUTH_IN                                | 18h   |
| AUTH_OUT                               | 19h   |

### 7.2.1 Authentication IN

This control request is used by a USB Host to transfer an Authentication Message without payload to a USB Device. This request initiates a control IN transfer.

The *wValue* and *wIndex* fields of an Authentication IN control packet contain the 4-byte Authentication Message Header described in Section 5.1. Table 7-5 shows the mapping of an Authentication Message Header onto the *wValue* and *wIndex* fields.

**Table 7-4: Authentication IN Control Request Fields**

| bmRequestType | bRequest | wValue                   | wIndex  | wLength | Data   |
|---------------|----------|--------------------------|---------|---------|--------|
| 10000000b     | AUTH_IN  | Authentication<br>Header | Message | Varies  | Varies |

**Table 7-5: Authentication Message Header Mapping**

| Offset | Field                  | Size | Value  | Description                        |
|--------|------------------------|------|--------|------------------------------------|
| 0      | <i>ProtocolVersion</i> | 1    | Number | Maps to high byte of <i>wValue</i> |
| 1      | <i>MessageType</i>     | 1    | Number | Maps to low byte of <i>wValue</i>  |
| 2      | <i>Param1</i>          | 1    | Number | Maps to high byte of <i>wIndex</i> |
| 3      | <i>Param2</i>          | 1    | Number | Maps to low byte of <i>wIndex</i>  |

The behavior of a USB Device is not specified if *wLength*, *wIndex*, or *wValue* for a particular Response type do not match the values set forth in this section. Otherwise, the following behavior is expected for the given USB Device state:

Default                      Not specified.

Address Request is valid and Device shall respond.  
Configured Request is invalid and Device shall respond with a Request Error.

### 7.2.2 Authentication OUT

This control request is used by a USB Host to transfer an Authentication Message with payload to a USB Device. This request initiates a control OUT transfer.

**Table 7-6: Authentication OUT Control Request Fields**

| bmRequestType | bRequest | wValue                | wIndex  | wLength | Data   |
|---------------|----------|-----------------------|---------|---------|--------|
| 00000000b     | AUTH_OUT | Authentication Header | Message | Varies  | Varies |

The *wValue* and *wIndex* fields of an Authentication OUT control packet contain the 4-byte Authentication Message Header described in Section 5.1. Table 7-5 shows the mapping of an Authentication Message Header onto the *wValue* and *wIndex* fields.

The behavior of a USB Device is not specified if *wLength*, *wIndex*, or *wValue* for a particular Authentication Request type do not match the values set forth in this section. Otherwise, the following behavior is expected for the given USB Device state:

Default Not specified.  
Address Request is valid and Device shall respond.  
Configured Request is invalid and Device shall respond with a Request Error.

## 7.3 Authentication Protocol

A USB Host can perform the Authentication operations described in Section 4. This section describes how each of the operations are performed.

### 7.3.1 Digest Query

A USB Host uses the GET\_DIGESTS Request defined in Section 5.2.1 to retrieve Certificate Chain digests from a USB Device. To send a GET\_DIGESTS Request, a USB Host initiates an AUTH\_IN Control Transfer with the values shown in Table 7-7.

**Table 7-7: GET\_DIGESTS Authentication IN Control Request Fields**

| bmRequestType | bRequest | wValue                                              | wIndex | wLength | Data             |
|---------------|----------|-----------------------------------------------------|--------|---------|------------------|
| 10000000b     | AUTH_IN  | See Table 5-4 for values and Table 7-5 for mapping. |        | 260     | DIGESTS Response |

In response to a GET\_DIGESTS Request, a USB Device returns either a DIGESTS Response as described in 5.3.1 or ERROR Response as described in Section 5.3.4.

### 7.3.2 Certificate Read

A USB Host uses an AUTH\_OUT control transfer followed by an AUTH\_IN control transfer to read a Certificate Chain or part thereof from a USB Device. The AUTH\_OUT control transfer contains a GET\_CERTIFICATE Request as shown in Table 7-8.

**Table 7-8: GET\_CERTIFICATE Authentication OUT Control Request Fields**

| bmRequestType | bRequest | wValue                                              | wIndex | wLength | Data          |
|---------------|----------|-----------------------------------------------------|--------|---------|---------------|
| 00000000b     | AUTH_OUT | See Table 5-5 for values and Table 7-5 for mapping. |        | 4       | See Table 5-6 |

After successfully completing the AUTH\_OUT control transfer, the USB Host initiates an AUTH\_IN Control transfer to retrieve the CERTIFICATE Response from the USB Device. The AUTH\_IN Control transfer contains the values in Table 7-9 where the high byte of *wValue* contains the *ProtocolVersion* field, the low byte of *wValue* contains the *MessageType* field in Table 5-12, and the *wLength* field contains the *length* value in the GET\_CERTIFICATE Response contained in the previous AUTH\_OUT Control transfer plus 4 (to account for the Authentication Message Header).

**Table 7-9: CERTIFICATE Authentication IN Control Request Fields**

| bmRequestType | bRequest | wValue         | wIndex | wLength    | Data                 |
|---------------|----------|----------------|--------|------------|----------------------|
| 10000000b     | AUTH_IN  | See Table 5-12 | 0      | length + 4 | CERTIFICATE Response |

In response to the AUTH\_IN Control transfer above, a USB Device returns either a CERTIFICATE Response as described in Section 5.3.2 or ERROR Response as described in Section 5.3.4.

### 7.3.3 Authentication Challenge

A USB Host uses an AUTH\_OUT control transfer followed by an AUTH\_IN control transfer to authenticate a USB Device. The AUTH\_OUT control transfer contains a CHALLENGE Request as shown in Table 7-10.

**Table 7-10: CHALLENGE Authentication OUT Control Request Fields**

| bmRequestType | bRequest | wValue                                              | wIndex | wLength | Data          |
|---------------|----------|-----------------------------------------------------|--------|---------|---------------|
| 00000000b     | AUTH_OUT | See Table 5-7 for values and Table 7-5 for mapping. |        | 32      | See Table 5-8 |

After successfully completing the AUTH\_OUT control transfer, the USB Host initiates an AUTH\_IN control transfer to retrieve the CHALLENGE\_AUTH Response from the USB Device. The AUTH\_IN control transfer contains the values in Table 7-11 where the high byte of *wValue* contains the *ProtocolVersion* field and the low byte of *wValue* contains the *MessageType* field in Table 5-14.

**Table 7-11: CHALLENGE\_AUTH Authentication IN Control Request Fields**

| bmRequestType | bRequest | wValue         | wIndex | wLength | Data                    |
|---------------|----------|----------------|--------|---------|-------------------------|
| 10000000b     | AUTH_IN  | See Table 5-14 | 0      | 168     | CHALLENGE_AUTH Response |

In response to the AUTH\_IN control transfer above, a USB Device returns either a CHALLENGE\_AUTH Response as described in Section 5.3.3 or ERROR Response as described in Section 5.3.4.

#### 7.3.4 Errors

If a USB Device encounters an Authentication-related error condition during an AUTH\_IN control transfer, it shall respond with an ERROR Response. If a USB Device encounters an Authentication-related error condition during an AUTH\_OUT control transfer, it shall respond to the next AUTH\_IN control transfer with an ERROR Response. The format for an ERROR Response is defined in Section 5.3.4.

#### 7.4 Timing Requirements for USB

All Authentication Message exchanges over USB shall follow the timing for control transfers set forth in **USB2.0** and **USB3.1**. Additional timing requirements are defined in Section 7.4.1 for USB Hosts and Section 7.4.2 for USB Devices.

##### 7.4.1 USB Host Timing Requirements

A USB Host shall use the timeout values in Table 7-12.

**Table 7-12: Authentication Initiator Timeout Values**

| Parameter            | Timeout Value | Description                                                                                 |
|----------------------|---------------|---------------------------------------------------------------------------------------------|
| <i>tDigestIN</i>     | 100 ms        | Timeout for a GET_DIGESTS Authentication Request.                                           |
| <i>tCertOUT</i>      | 100 ms        | Timeout for an AUTH_OUT control transfer carrying a GET_CERTIFICATE Authentication Request. |
| <i>tCertIN</i>       | 500 ms        | Timeout for an AUTH_IN control transfer carrying a CERTIFICATE Authentication Response.     |
| <i>tChallengeOUT</i> | 100 ms        | Timeout for an AUTH_OUT control transfer carrying a CHALLENGE Authentication Request.       |
| <i>tChallengeIN</i>  | 600 ms        | Timeout for an AUTH_IN control transfer carrying a CHALLENGE_AUTH Authentication Response.  |

If a USB Host does not receive an Authentication Response within  $t_{DigestIN}$  of sending a GET\_DIGESTS Authentication Request, it is considered an error. Subsequent handling of this error is outside the scope of this specification.

If a USB Host does not receive an ACK within  $t_{CertOUT}$  of sending a GET\_CERTIFICATE Authentication Request to an Authentication Responder, it is considered an error. Subsequent handling of this error is outside the scope of this specification.

If a USB Host does not receive an Authentication Response within  $t_{CertIN}$  of sending a CERTIFICATE Authentication Request to an Authentication Responder, it is considered an error. Subsequent handling of this error is outside the scope of this specification.

If a USB Host does not receive an ACK within  $t_{ChallengeOUT}$  of sending a CHALLENGE Authentication Request to an Authentication Responder, it is considered an error. Subsequent handling of this error is outside the scope of this specification.

If a USB Host does not receive an Authentication Response within  $t_{ChallengeIN}$  of sending a CHALLENGE\_AUTH Authentication Request to an Authentication Responder, it is considered an error. Subsequent handling of this error is outside the scope of this specification.

#### 7.4.2 USB Device Timing Requirements

A USB Device shall use the response times in Table 7-13.

**Table 7-13: Authentication Responder Response Times**

| Parameter               | Maximum Response Time | Description                                                                                                    |
|-------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------|
| $t_{DigestSent}$        | 95 ms                 | Maximum time between receiving a DIGESTS Authentication Request and sending an Authentication Response.        |
| $t_{CertACK}$           | 95 ms                 | Maximum time between receiving a GET_CERTIFICATE Request and sending an ACK.                                   |
| $t_{CertSent}$          | 495 ms                | Maximum time between receiving a CERTIFICATE Authentication Request and sending an Authentication Response.    |
| $t_{ChallengeACK}$      | 95 ms                 | Maximum time between receiving a CHALLENGE Request and sending an ACK.                                         |
| $t_{ChallengeAuthSent}$ | 595 ms                | Maximum time between receiving a CHALLENGE_AUTH Authentication Request and sending an Authentication Response. |

A USB Device shall respond to an Authentication Initiator within *tDigestSent* of receiving an AUTH\_IN control transfer carrying a GET\_DIGESTS Authentication Request.

A USB Device shall ACK an AUTH\_OUT control transfer carrying a GET\_CERTIFICATE Request within *tCertACK* of receiving it.

A USB Device shall respond to an Authentication Initiator within *tCertSent* of receiving an AUTH\_IN control transfer carrying a CERTIFICATE Authentication Request.

A USB Device shall ACK an AUTH\_OUT control transfer carrying a CHALLENGE Request within *tChallengeACK* of receiving it.

A USB Device shall respond to an Authentication Initiator within *tChallengeAuthSent* of receiving an AUTH\_IN control transfer carrying a CHALLENGE\_AUTH Authentication Request.

## 7.5 Context Hash

This field shall contain a 32-byte **SHA256** hash of the following USB Descriptor data (as defined in **USB2.0** and **USB3.1**) for current operating speed, concatenated together in following order:

1. Device Descriptor
2. Complete BOS Descriptor (if present)
3. Complete Configuration 1 Descriptor
4. Complete Configuration 2 Descriptor (if present)
5. ...
6. Complete Configuration n Descriptor (if present)

The contents of each descriptor listed above shall match that which the device presents during enumeration at the USB Device's current connection.

## 8 Protocol Constants

Table 8-1: Protocol Constants

| Constant                       | Value      |
|--------------------------------|------------|
| <i>MaxLeafCertSize</i>         | 640 bytes  |
| <i>MaxIntermediateCertSize</i> | 512 bytes  |
| <i>MaxACDSize</i>              | 128 bytes  |
| <i>MaxCertChainSize</i>        | 4096 bytes |



## A ACD

### A.1. ACD Formatting

ACD formatting consists of a sequence of zero or more Type, Length, Value (TLV) fields that start at the first byte of a binary object. The general format for a TLV field is set out in Table A-1. TLV types are listed in Table A-2 and defined in more detail below.

Each TLV type shall occur no more than once. Each TLV shall appear in increasing order by TLV value.

All TLVs are big-endian (i.e. MSB to LSB).

**Table A-1: TLV General Format**

| Offset | Field  | Size   | Description                   |
|--------|--------|--------|-------------------------------|
| 0      | Type   | 1      | TLV Type                      |
| 1      | Length | 1      | Number of bytes in Data field |
| 2      | Data   | Length | Determined by TLV type        |

**Table A-2: TLV Types**

| Value     | Name                        |
|-----------|-----------------------------|
| 00h       | VERSION                     |
| 01h       | XID                         |
| 02h       | POWER_SOURCE_CAPABILITIES   |
| 03h       | POWER_SOURCE_CERTIFICATIONS |
| 04h       | CABLE_CAPABILITIES          |
| 05h       | SECURITY_DESCRIPTION        |
| 06h - FCh | Reserved                    |
| FDh       | PLAYPEN                     |
| FEh       | VENDOR_EXTENSION            |
| FFh       | EXTENSION                   |

#### A.1.1. Version TLV

This TLV is used to specify the ACD type and version. Valid ACD Versions are listed in Table A-4.

**Table A-3: Version TLV Fields**

| Offset | Field  | Size | Value          |
|--------|--------|------|----------------|
| 0      | Type   | 1    | VERSION        |
| 1      | Length | 1    | 2              |
| 2      | Data   | 2    | See Figure A-1 |

**Figure A-1: Bitmap of Version TLV Data**

| Bit[15]     | Bit[14] | Bit[13] | Bit[12]  | Bit[11] | Bit[10] | Bit[9] | Bit[8] |
|-------------|---------|---------|----------|---------|---------|--------|--------|
| USB*        | PD*     | Cable*  | Reserved |         |         |        |        |
| Bit[7]      | Bit[6]  | Bit[5]  | Bit[4]   | Bit[3]  | Bit[2]  | Bit[1] | Bit[0] |
| ACD Version |         |         |          |         |         |        |        |

\* Bit 15 is set to 1 for a USB Product ACD. Bit 14 is set to 1 for a PD Product ACD. Bit 13 is set to 1 for a USB Type-C Cable.

**Table A-4: ACD Version Encoding**

| Value   | Description                                                                               |
|---------|-------------------------------------------------------------------------------------------|
| 0       | Adheres to the ACD format defined in USB Type-C Authentication Specification Revision 1.0 |
| 1 - 255 | Reserved                                                                                  |

**A.1.2. XID TLV**

This TLV is used to convey the 32-bit XID originating from the USB-IF for the purposes of compliance testing ([http://www.usb.org/developers/compliance/request\\_XID/](http://www.usb.org/developers/compliance/request_XID/)).

**Table A-5: XID TLV Fields**

| Offset | Field  | Size | Value                                                                |
|--------|--------|------|----------------------------------------------------------------------|
| 0      | Type   | 1    | XID                                                                  |
| 1      | Length | 1    | 4                                                                    |
| 2      | Data   | 4    | Vendor-selected from a block of 32-bit values assigned by the USB-IF |

**A.1.3. Power Source Capabilities TLV**

This TLV is used to specify the power source capabilities of a PD power source.

**Table A-6: Power Source Capabilities TLV Fields**

| Offset | Field  | Size   | Value                     |
|--------|--------|--------|---------------------------|
| 0      | Type   | 1      | POWER_SOURCE_CAPABILITIES |
| 1      | Length | 1      | Varies                    |
| 2      | Data   | Length | See Table A-7             |

**Table A-7: Power Source Capabilities TLV Data**

| Offset | Field              | Size         | Description                                                                                                                                  |
|--------|--------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | Version            | 1            | Upper nibble contains the TLV Version<br>Lower nibble contains the PD Revision. See <b>USBPD</b> Section 6.2.1.1.5 (Specification Revision). |
| 1      | FW Version         | 1            | See <b>USBPD</b> , Section 6.5.1.2 (Product ID Field) and Table 6-37 (Source Capabilities Extended Data Block).                              |
| 2      | HW Version         | 1            | See <b>USBPD</b> , Section 6.5.1.3 (Hardware Version Field).                                                                                 |
| 3      | Voltage Regulation | 1            | See <b>USBPD</b> , Section 6.5.1.4 (Voltage Regulation Field).                                                                               |
| 4      | Hold Up time       | 1            | See <b>USBPD</b> , Section 6.5.1.5 (Holdup Time Field).                                                                                      |
| 5      | Compliance         | 1            | See <b>USBPD</b> , Section 6.5.1.6 (Compliance Field).                                                                                       |
| 6      | Touch Current      | 1            | See <b>USBPD</b> , Section 6.5.1.7 (Touch Current).                                                                                          |
| 7      | Reserved           | 1            | Reserved.                                                                                                                                    |
| 8      | Peak Current 1     | 2            | See <b>USBPD</b> , Section 6.5.1.8 (Peak Current).                                                                                           |
| 10     | Peak Current 2     | 2            | See <b>USBPD</b> , Section 6.5.1.8 (Peak Current).                                                                                           |
| 12     | Peak Current 3     | 2            | See <b>USBPD</b> , Section 6.5.1.8 (Peak Current).                                                                                           |
| 14     | Touch Temp         | 1            | See <b>USBPD</b> , Section 6.5.1.9 (Touch Temp).                                                                                             |
| 15     | Source Inputs      | 1            | See <b>USBPD</b> , Section 6.5.1.10 (Source Inputs).                                                                                         |
| 16     | Batteries          | 1            | See <b>USBPD</b> , Section 6.5.1.11 (Batteries).                                                                                             |
| 17     | Num PDOs           | 1            | See <b>USBPD</b> , Section 6.2.1.1.2 (Number of Data Objects).                                                                               |
| 18     | PDOs               | Num PDOs * 4 | See <b>USBPD</b> , Section 6.4.1.2 (Source Capabilities Message).                                                                            |

#### A.1.4. Power Source Certifications TLV

This TLV contains a bitmap attesting to the certifications a PD Product has achieved.

**Table A-8: Power Source Certifications TLV Fields**

| Offset | Field  | Size   | Value                       |
|--------|--------|--------|-----------------------------|
| 0      | Type   | 1      | POWER_SOURCE_CERTIFICATIONS |
| 1      | Length | 1      | Varies                      |
| 2      | Data   | Length | Reserved                    |

**A.1.5. Cable Capabilities TLV****Table A-9: Cable Capabilities TLV Fields**

| Offset | Field  | Size | Value              |
|--------|--------|------|--------------------|
| 0      | Type   | 1    | CABLE_CAPABILITIES |
| 1      | Length | 1    | 6                  |
| 2      | Data   | 6    | See Table A-10     |

**Table A-10: Cable Capabilities TLV Data**

| Offset | Field        | Size | Description                                                                                                                                    |
|--------|--------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | Version      | 1    | Upper nibble contains the TLV Version<br>Lower nibble contains the PD Revision. See <b>USBPD</b> , Section 6.2.1.1.5 (Specification Revision). |
| 1      | Product Type | 1    | See <b>USBPD</b> , Section 6.4.4.3.1.1.4 (Product Type Cable Plug).                                                                            |
| 2      | Cable VDO    | 4    | See <b>USBPD</b> , Table 6-30 (Passive Cable VDO) and Table 6-31 (Active Cable VDO). See also <b>USBPD</b> Section 6.4.4.3.1.2 (Cable VDO).    |

**A.1.6. Security Description TLV**

This TLV contains information about the secure components of a Product and their trustworthiness. The certifications claimed in this field shall be relevant and appropriate to the security functions used in the product.

**Table A-11: Security Description TLV Fields**

| Offset | Field  | Size | Value                |
|--------|--------|------|----------------------|
| 0      | Type   | 1    | SECURITY_DESCRIPTION |
| 1      | Length | 1    | 6                    |
| 2      | Data   | 6    | See Table A-12       |

**Table A-12: Security Data**

| Offset | Field                        | Size | Value               |
|--------|------------------------------|------|---------------------|
| 0      | FIPS/ISO Identifier          | 1    | See Section A.1.6.1 |
| 1      | Common Criteria Identifier   | 2    | See Section A.1.6.2 |
| 3      | Security Analysis Identifier | 1    | See Section A.1.6.3 |
| 4      | IC Vendor Identifier         | 2    | See Section A.1.6.4 |

**A.1.6.1. FIPS/ISO Identifier**

This field encodes the NIST-FIPS-140-2 or ISO-19790 security level of the Product. A Product uses the values listed in Table A-13 to indicate which certification it has received (if any).

**Table A-13: FIPS/ISO Level Identifiers**

| Value    | Description                                     |
|----------|-------------------------------------------------|
| 0        | No FIPS/ISO certification                       |
| 1        | ISO-19790 (ed1: 2006), NIST-FIPS 140-2, Level 1 |
| 2        | ISO-19790 (ed1: 2006), NIST-FIPS 140-2, Level 2 |
| 3        | ISO-19790 (ed1: 2006), NIST-FIPS 140-2, Level 3 |
| 4        | ISO-19790 (ed1: 2006), NIST-FIPS 140-2, Level 4 |
| 5-8      | Reserved                                        |
| 9        | ISO-19790 (ed2: 2012), Level 1                  |
| 10       | ISO-19790 (ed2: 2012), Level 2                  |
| 11       | ISO-19790 (ed2: 2012), Level 3                  |
| 12       | ISO-19790 (ed2: 2012), Level 4                  |
| 13 - 255 | Reserved                                        |

#### A.1.6.2. Common Criteria Identifier

This field encodes **Common Criteria** information for a product in the format shown in Figure A-1.

**Figure A-2: Bitmap of the Common Criteria Identifier**

| Bit[15]                     | Bit[14] | Bit[13]   | Bit[12]            | Bit[11] | Bit[10]                  | Bit[9] | Bit[8] |
|-----------------------------|---------|-----------|--------------------|---------|--------------------------|--------|--------|
| Development Security        |         | CM*       | Certification Year |         |                          |        |        |
| Bit[7]                      | Bit[6]  | Bit[5]    | Bit[4]             | Bit[3]  | Bit[2]                   | Bit[1] | Bit[0] |
| Protection Profile Encoding |         | EAL Level |                    |         | Vulnerability Assessment |        |        |

\* CM: Certificate Maintenance

#### A.1.6.2.1 Vulnerability Assessment

The vulnerability assessment (AVA\_VAN) certifies a resistance to attackers with certain attack potential (AVA\_VAN Definition: “Common Criteria for Information Technology Security Evaluation, Part 3: Security assurance components (September 2012, Version 3.1, Revision 4)”, <https://www.commoncriteriaportal.org/files/ccfiles/CCPART3V3.1R4.pdf>, see “8 Evaluation assurance levels” and “16 Class AVA: Vulnerability assessment”). USB Products should achieve AVA\_VAN.5 to provide a sufficient level of protection. Vulnerability assessment encodings are shown in Table A-14.

**Table A-14: Vulnerability Assessment**

| Value | Description                                          |
|-------|------------------------------------------------------|
| 0     | No vulnerability assessment performed                |
| 1     | AVA_VAN.1 Vulnerability survey                       |
| 2     | AVA_VAN.2 Vulnerability analysis                     |
| 3     | AVA_VAN.3 Focused vulnerability analysis             |
| 4     | AVA_VAN.4 Methodical vulnerability analysis          |
| 5     | AVA_VAN.5 Advanced methodical vulnerability analysis |

#### A.1.6.2.2 EAL Level

The EAL level defines the Evaluation Assurance Level (EAL) of the USB Product and indicates the confidence and amount of testing performed by the certification agency. EAL levels encodings are shown in Table A-15.

**Table A-15: EAL Encodings**

| Value | Description                                       |
|-------|---------------------------------------------------|
| 0     | No <b>Common Criteria</b> certification performed |
| 1     | EAL1: Functionally Tested                         |
| 2     | EAL2: Structurally Tested                         |
| 3     | EAL3: Methodically Tested and Checked             |
| 4     | EAL4: Methodically Designed, Tested and Reviewed  |
| 5     | EAL5: Semi-formally Designed and Tested           |
| 6     | EAL6: Semi-formally Verified Design and Tested    |
| 7     | EAL7: Formally Verified Design and Tested         |

#### A.1.6.2.3 Protection Profile

An appropriate protection profile for **Common Criteria** evaluation should be used. The list of appropriate protection profiles is available on the Common Criteria Portal (available at: <https://www.commoncriteriaportal.org/pps/>). The two protection profiles for “ICs, Smart Cards and Smart-card related devices and systems” are: BSI-CC-PP-0084-2014 and BSI-PP-0035-2007. The different encodings for the Protection Profile field are provided in Table A-16.

**Table A-16: Protection Profile Encoding**

| Value | Description                                                 |
|-------|-------------------------------------------------------------|
| 0     | No <b>Common Criteria</b> evaluation performed              |
| 1     | Security IC platform protection profile BSI-PP-0035-2007    |
| 2     | Security IC platform protection profile BSI-CC-PP-0084-2014 |
| 3     | Other protection profile listed in <b>Common Criteria</b>   |

#### A.1.6.2.4 Development Security

The life cycle support development security (ALC\_DVS) is concerned with physical, procedural, personnel, and other security measures that may be used in the development environment to protect the TOE (Target of Evaluation). It includes the physical security of the development location(s) and controls on the selection and hiring of development staff. (ALC\_DVS Definition: “Common Criteria for Information Technology Security Evaluation, Part 3: Security assurance components (September 2012, Version 3.1, Revision 4)”, <https://www.commoncriteriaportal.org/files/ccfiles/CCPART3V3.1R4.pdf>, see “14.4 Development Security”).

**Table A-17: Development Security**

| Value | Description                                         |
|-------|-----------------------------------------------------|
| 0     | No ALC_DVS compliant development security performed |
| 1     | ALC_DVS.1 Identification of Security Measures       |
| 2     | ALC_DVS.2 Sufficiency of Security Measures          |
| 3     | Reserved                                            |

#### **A.1.6.2.5 Certification Maintenance**

Declares that, at the time of the corresponding Product's production completion date, the specified **Common Criteria** certification with augmentations was still valid and in force, from either the original certification or extended through the use of certification maintenance packages accepted by **Common Criteria** for the original certification.

**Table A-18: Certification Maintenance**

| Value | Description                                                                                   |
|-------|-----------------------------------------------------------------------------------------------|
| 0     | No Declaration of <b>Common Criteria</b> certification validity at production completion date |
| 1     | <b>Common Criteria</b> certification valid at Product's production completion date            |

#### **A.1.6.2.6 Certification Year**

Specifies the year that the corresponding **Common Criteria** certificate was obtained. The year value is specified as (year – 2010) and encoded into 5 bits. This allows for certification years in the range from 2010 to 2041.

#### **A.1.6.3. Security Analysis Identifier**

This field expresses the degree of attack resistance that was established outside of the FIPS or **Common Criteria** regime either by internal testing carried out by the vendor (or its supplier), testing carried out by an external lab, or by deriving the degree of assurance from an alternative or future certification. The level of attack resistance is measured according to the rating in the JIL/JHAS document "Application of Attack Potential to Smartcards" (JIL/JHAS: "Application of Attack Potential to Smartcards - Joint Interpretation Library (Version 2.9, January 2013)", <http://www.sogisportal.eu/documents/cc/domains/sc/JIL-Application-of-Attack-Potential-to-Smartcards-v2-9.pdf>). The encoding is described in Figure A-2.

**Figure A-3: Bitmap of the Security Analysis Identifier**

| Bit[7]   | Bit[6]              | Bit[5] | Bit[4] | Bit[3]         | Bit[2] | Bit[1] | Bit[0] |
|----------|---------------------|--------|--------|----------------|--------|--------|--------|
| Reserved | JHAS/JIL resistance |        |        | Testing method |        |        |        |

#### A.1.6.3.1 Testing Method

A vendor can use the Bit fields shown in Table A-19 to express how the stated level of attack resistance was determined. Possible options are internal testing, testing by an external lab, or the derivation of the attack resistance from a certification. This can either be **Common Criteria**, or FIPS, but also different or future security certification and evaluation standards. It is possible to set multiple bits, e.g., if internal testing, external testing, and certification was performed. If no bits are set, then no testing or security evaluation is claimed.

**Table A-19: Testing Method Encoding**

| Bit    | Description                                                                  |
|--------|------------------------------------------------------------------------------|
| Bit[0] | Internally tested by vendor                                                  |
| Bit[1] | Tested by external lab                                                       |
| Bit[2] | Derived from certification (not exclusive to <b>Common Criteria</b> or FIPS) |
| Bit[3] | Reserved                                                                     |

#### A.1.6.3.2 JIL/JHAS Resistance

The encodings for the resistance against attackers with a certain potential are shown in Table A-20. The rating is based on the JIL/JHAS document “Application of Attack Potential to Smartcards” (JIL/JHAS: “Application of Attack Potential to Smartcards - Joint Interpretation Library (Version 2.9, January 2013)”, <http://www.sogisportal.eu/documents/cc/domains/sc/JIL-Application-of-Attack-Potential-to-Smartcards-v2-9.pdf>) and in case this document was not used for evaluation or internal testing it is up to the vendor to rate his product appropriately.

**Table A-20: Vulnerability Assessment**

| Value | Description                           |
|-------|---------------------------------------|
| 0     | No vulnerability assessment performed |
| 1     | No rating                             |
| 2     | Basic                                 |
| 3     | Enhanced-Basic                        |
| 4     | Moderate                              |
| 5     | High                                  |
| 6 -7  | Reserved                              |

#### A.1.6.4. IC Vendor Identifier

This 2-byte field shall contain either the USB-IF-assigned VID that identifies the IC vendor or zero if not used.

#### A.1.7. Playpen TLV

This TLV is used for development purposes only. It shall not be used or interpreted by any Products.



**Table A-21: Playpen TLV Fields**

| Offset | Field  | Size   | Value                  |
|--------|--------|--------|------------------------|
| 0      | Type   | 1      | PLAYPEN                |
| 1      | Length | 1      | To be set by developer |
| 2      | Data   | Length | To be set by developer |

#### **A.1.8. Vendor Extension TLV**

This TLV contains a vendor proprietary information. The first two bytes of the Data field shall always contain the Vendor ID of the vendor defining the field.

**Table A-22: Vendor Extension TLV Fields**

| Offset | Field  | Size   | Value            |
|--------|--------|--------|------------------|
| 0      | Type   | 1      | VENDOR_EXTENSION |
| 1      | Length | 1      | Vendor defined   |
| 2      | Data   | Length | See Table A-23   |

**Table A-23: Vendor Extension TLV Data**

| Offset | Field          | Size         | Description                                 |
|--------|----------------|--------------|---------------------------------------------|
| 0      | VID            | 2            | Vendor ID of vendor defining the Data field |
| 2      | Vendor Defined | (Length – 2) | Vendor defined                              |

#### **A.1.9. Extension TLV**

This TLV is used to address future exhaustion of the TLV space. There are currently no Extension types and this TLV type shall not be used.

**Table A-24: Extension TLV Fields**

| Offset | Field  | Size   | Value                |
|--------|--------|--------|----------------------|
| 0      | Type   | 1      | EXTENSION            |
| 1      | Length | 1      | Length of Data field |
| 2      | Data   | Length | Extension type data  |

#### **A.2. ACD for a PD Product**

The ACD for a PD Product is defined in Table A-25. All TLV Types marked as “Required” shall be present. Types marked “N/A” are not allowed and shall not be used.

**Table A-25: PD Product ACD TLVs**

| Value     | Name                        | PD Source/Sink | USB Type-C Cable |
|-----------|-----------------------------|----------------|------------------|
| 00h       | VERSION                     | Required       | Required         |
| 01h       | XID                         | Required       | Required         |
| 02h       | POWER_SOURCE_CAPABILITIES   | Required       | N/A              |
| 03h       | POWER_SOURCE_CERTIFICATIONS | Reserved       | N/A              |
| 04h       | CABLE_CAPABILITIES          | N/A            | Required         |
| 05h       | SECURITY_DESCRIPTION        | Required       | Required         |
| 06h - FCh | Reserved                    | --             | --               |
| FDh       | PLAYPEN                     | Optional       | Optional         |
| FEh       | VENDOR_EXTENSION            | Optional       | Optional         |
| FFh       | EXTENSION                   | Optional       | Optional         |

### A.3. ACD for a USB Product

The ACD for a USB Product is defined in Table A-26. All TLV Types marked as “Required” shall be present. Types marked “N/A” are not allowed and shall not be used. Types marked as “Conditional” may only be used in PDUSB Products.

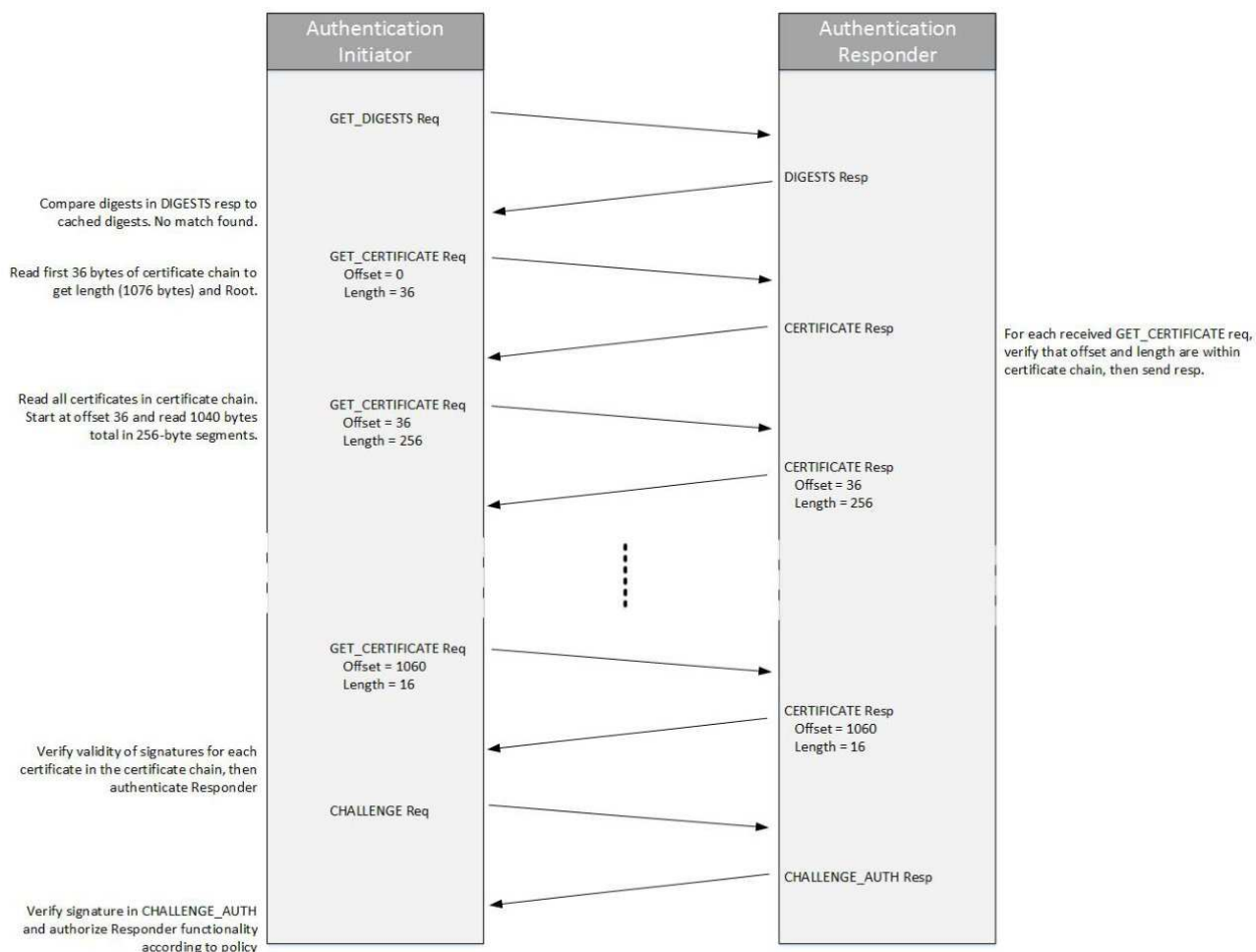
**Table A-26: USB Product ACD TLVs**

| Value     | Name                        | USB         |
|-----------|-----------------------------|-------------|
| 00h       | VERSION                     | Required    |
| 01h       | XID                         | Optional    |
| 02h       | POWER_SOURCE_CAPABILITIES   | Conditional |
| 03h       | POWER_SOURCE_CERTIFICATIONS | Conditional |
| 04h       | CABLE_CAPABILITIES          | N/A         |
| 05h       | SECURITY_DESCRIPTION        | Required    |
| 06h - FCh | Reserved                    | --          |
| FDh       | PLAYPEN                     | Optional    |
| FEh       | VENDOR_EXTENSION            | Optional    |
| FFh       | EXTENSION                   | Optional    |

## B Cryptographic Examples

The examples in this Appendix are included to help illustrate the concepts defined in the specification and are informative only.

### B.1. Example Authentication Sequence



### B.2. Example Certificate Chain Topology

#### B.2.1. Certificate Chain

```

87 03 00 00 EB 13 EB C1 8D F6 73 03 9B 76 99 66
AD A3 E5 26 AC 40 77 09 C2 37 24 FB E0 C7 B2 E0
02 30 FF 69 30 82 01 80 30 82 01 25 A0 03 02 01
02 02 09 00 83 74 6A 3F AE 6A DF 28 30 0A 06 08
2A 86 48 CE 3D 04 03 02 30 24 31 12 30 10 06 03
55 04 0A 0C 09 55 53 42 2D 49 46 20 43 41 31 0E
  
```

```

30 0C 06 03 55 04 03 0C 05 55 53 42 3A 3A 30 22
18 0F 31 39 37 30 30 31 30 31 30 30 30 30 30
5A 18 0F 39 39 39 39 31 32 33 31 32 33 35 39 35
39 5A 30 2B 31 15 30 13 06 03 55 04 0A 0C 0C 4F
72 67 4E 61 6D 65 20 49 6E 63 2E 31 12 30 10 06
03 55 04 03 0C 09 55 53 42 3A 31 61 30 61 3A 30
59 30 13 06 07 2A 86 48 CE 3D 02 01 06 08 2A 86
48 CE 3D 03 01 07 03 42 00 04 3F 11 11 AC 9A A1
C2 43 8E 66 D8 41 C1 4F 87 6A DF 35 0E AA 73 6C
89 44 5F 42 19 B3 5C F8 F3 37 C8 9D 64 EC 65 B1
16 4D C7 E6 89 60 62 6A D5 D5 AD 37 21 D3 60 39
38 1B 2F 01 4B 43 09 4C 85 B1 A3 35 30 33 30 0F
06 03 55 1D 13 01 01 FF 04 05 30 03 01 01 FF 30
0B 06 03 55 1D 0F 04 04 03 02 01 06 30 13 06 03
55 1D 25 01 01 FF 04 09 30 07 06 05 67 81 11 01
01 30 0A 06 08 2A 86 48 CE 3D 04 03 02 03 49 00
30 46 02 21 00 9D F3 20 EC 82 63 F4 D0 74 1E D5
45 A0 B8 D5 FA 07 59 42 88 DE 82 BA 46 A9 D0 91
09 41 CC 6D A6 02 21 00 C3 CC 17 0C B5 AF 5D F3
A5 53 81 D4 7D 7B 06 AD A3 0A A7 93 1C A7 B0 11
6A D2 0D 3C 16 94 95 7B 30 82 01 DB 30 82 01 81
A0 03 02 01 02 02 09 00 81 9F 59 97 6B 38 47 91
30 0A 06 08 2A 86 48 CE 3D 04 03 02 30 2B 31 15
30 13 06 03 55 04 0A 0C 0C 4F 72 67 4E 61 6D 65
20 49 6E 63 2E 31 12 30 10 06 03 55 04 03 0C 09
55 53 42 3A 31 61 30 61 3A 30 22 18 0F 31 39 37
30 30 31 30 31 30 30 30 30 30 30 5A 18 0F 39 39
39 39 31 32 33 31 32 33 35 39 35 39 5A 30 42 31
15 30 13 06 03 55 04 0A 0C 0C 4F 72 67 4E 61 6D
65 20 49 6E 63 2E 31 16 30 14 06 03 55 04 03 0C
0D 55 53 42 3A 31 61 30 61 3A 30 31 30 31 31 11
30 0F 06 03 55 04 05 13 08 35 35 36 36 37 37 38
38 30 59 30 13 06 07 2A 86 48 CE 3D 02 01 06 08
2A 86 48 CE 3D 03 01 07 03 42 00 04 A9 B0 F8 66
B0 2D 91 2B 87 64 22 57 D2 AE 0B 07 E1 FA 83 A6
8E B4 4F 3F 16 79 43 A3 E5 D8 00 72 DC 0A D5 B3
00 A4 FB 8B C0 05 3B 4D 7C 9D 8D 48 BB AC 46 8C
E5 28 B7 5B 1C 5A FA 2E 4D DA 38 45 A3 73 30 71
30 0C 06 03 55 1D 13 01 01 FF 04 02 30 00 30 0B
06 03 55 1D 0F 04 04 03 02 07 80 30 13 06 03 55
1D 25 01 01 FF 04 09 30 07 06 05 67 81 11 01 01
30 3F 06 05 67 81 11 01 02 04 36 00 02 40 00 01
04 00 00 12 34 02 16 02 01 01 00 03 07 01 00 2A
0A 2A 0A 2A 0A 00 00 00 01 2A 01 91 2C 05 06 00
00 00 55 1A 0A FD 04 54 45 53 54 FE 04 1A 0A 12
34 30 0A 06 08 2A 86 48 CE 3D 04 03 02 03 48 00
30 45 02 20 73 DB 58 34 78 91 0C B9 C3 F5 6C 00
AB 6E 9C E0 E0 13 B1 A3 41 D4 12 5F 3B E0 E9 31
A3 C3 6E D5 02 21 00 C6 CF 1D BD 9D 0F B1 2C 5B
EF 17 4E 30 44 52 C3 27 FA A2 FA 40 29 F5 49 65
31 21 05 C2 BB 2B 35

```

#### B.2.1.1. Intermediate Certificate

```
-----BEGIN CERTIFICATE-----
MIIBgDCCASWgAwIBAgIJAIN0aj+uat8oMAoGCCqGSM49BAMCMCQxEjAQBgNVBAoM
CVVVTQi1JRiBDQTEOMAwGA1UEAwwFVVNCOjowIhgPMTk3MDAxMDEwMDAwMDBaGA85
OTk5MTIzMTIzNTk1OVowKzEVMBMGA1UECgwMT3JnTmFtZSBjbmMuMURiEAYDVQQD
DA1VU0I6MWEwYTowWTATBgcqhkiOPOQIBBgqhkiOPOQMBBwNCAAQ/ERGsmqHCQ45m
2EHBT4dq3zUOqnNsiURfQhmzXPjzN8idZOxlsRZNx+aJYGJq1dWtNyHTYDk4Gy8B
S0MJTIWxozUwMzAPBgNVHRMBAf8EBTADAQH/MAsGA1UdDwQEAwIBBjATBgNVHSUB
Af8ECTAHBgVngREBATAKBggqhkiOPOQDDAgNjADBGAiEAnfMg7IJj9NB0HtVFoLjV
+gdZQojegrpGqdCRCUHMbaYCIQDDZBcMta9d86VTgdR9ewatowqnkxynsBFq0g08
FpSVew==
-----END CERTIFICATE-----
Certificate:
  Data:
    Version: 3 (0x2)
    Serial Number:
      83:74:6a:3f:ae:6a:df:28
    Signature Algorithm: ecdsa-with-SHA256
    Issuer: O=USB-IF CA, CN=USB::
    Validity
      Not Before: Jan  1 00:00:00 1970 GMT
      Not After : Dec 31 23:59:59 9999 GMT
    Subject: O=OrgName Inc., CN=USB:1a0a:
    Subject Public Key Info:
      Public Key Algorithm: id-ecPublicKey
      Public-Key: (256 bit)
      pub:
        04:3f:11:11:ac:9a:a1:c2:43:8e:66:d8:41:c1:4f:
        87:6a:df:35:0e:aa:73:6c:89:44:5f:42:19:b3:5c:
        f8:f3:37:c8:9d:64:ec:65:b1:16:4d:c7:e6:89:60:
        62:6a:d5:d5:ad:37:21:d3:60:39:38:1b:2f:01:4b:
        43:09:4c:85:b1
      ASN1 OID: prime256v1
      NIST CURVE: P-256
    X509v3 extensions:
      X509v3 Basic Constraints: critical
        CA:TRUE
      X509v3 Key Usage:
        Certificate Sign, CRL Sign
      X509v3 Extended Key Usage: critical
        2.23.145.1.1
    Signature Algorithm: ecdsa-with-SHA256
      30:46:02:21:00:9d:f3:20:ec:82:63:f4:d0:74:1e:d5:45:a0:
      b8:d5:fa:07:59:42:88:de:82:ba:46:a9:d0:91:09:41:cc:6d:
      a6:02:21:00:c3:cc:17:0c:b5:af:5d:f3:a5:53:81:d4:7d:7b:
      06:ad:a3:0a:a7:93:1c:a7:b0:11:6a:d2:0d:3c:16:94:95:7b
```

#### B.2.1.2. Leaf Certificate

```
-----BEGIN CERTIFICATE-----
MIIB2zCCAYGgAwIBAgIJAIgfWZdrOEeRMAoGCCqGSM49BAMCMCsxFTATBgNVBAoM
DE9yZ05hbWUgSW5jLjESMBAGA1UEAwwJVVNCOjFhMGE6MCIYDzE5NzAwMTAxMDAw
```

```
MDAwWhgPOTk5OTEyMzEyMzU5NTlaMEIxFtATBgNVBAoMDE9yZ05hbWUgSW5jLjEW
MBQGA1UEAwwNVVNCOjFhMGE6MDEwMTERMA8GA1UEBRMINTU2Njc3ODgwWTATBgccq
hkjOPQIBBggqhkhkjOPQMBBwNCAASpsPhmsC2RK4dkIlfSrgsH4fqDpo60Tz8WeUOj
5dgActwK1bMapPuLwAU7TXydjUi7rEaM5Si3Wxxa+i5N2jhFo3MwctAMBgNVHRMB
Af8EAjAAMAsGA1UdDwQEAwIHgDATBgNVHSUBAf8ECTAHBgVngREBATA/BgVngREB
AgQ2AAJAAAEAAAASNAIWAgEBAAMHAQAqCioKKgoAAAABKgGRLAUGAAAARoK/QRU
RVNU/gQaChI0MAoGCCqGSM49BAMCA0gAMEUCIHPbWDR4kQy5w/VsAKtunODgE7Gj
QdQsXzvg6TGjw27VAiEAXs8dvZ0PsSxb7xdOMERSwyf6ovpAKfVJZTEhBcK7KzU=
-----END CERTIFICATE-----
```

Certificate:

Data:

Version: 3 (0x2)

Serial Number:

81:9f:59:97:6b:38:47:91

Signature Algorithm: ecdsa-with-SHA256

Issuer: O=OrgName Inc., CN=USB:1a0a:

Validity

Not Before: Jan 1 00:00:00 1970 GMT

Not After : Dec 31 23:59:59 9999 GMT

Subject: O=OrgName

Inc.,

CN=USB:1a0a:0101/serialNumber=55667788

Subject Public Key Info:

Public Key Algorithm: id-ecPublicKey

Public-Key: (256 bit)

pub:

04:a9:b0:f8:66:b0:2d:91:2b:87:64:22:57:d2:ae:

0b:07:e1:fa:83:a6:8e:b4:4f:3f:16:79:43:a3:e5:

d8:00:72:dc:0a:d5:b3:00:a4:fb:8b:c0:05:3b:4d:

7c:9d:8d:48:bb:ac:46:8c:e5:28:b7:5b:1c:5a:fa:

2e:4d:da:38:45

ASN1 OID: prime256v1

NIST CURVE: P-256

X509v3 extensions:

X509v3 Basic Constraints: critical

CA:FALSE

X509v3 Key Usage:

Digital Signature

X509v3 Extended Key Usage: critical

2.23.145.1.1

2.23.145.1.2:

..@.....4.....\*

\*

\*

....\*...U.

..TEST...

.4

Signature Algorithm: ecdsa-with-SHA256

30:45:02:20:73:db:58:34:78:91:0c:b9:c3:f5:6c:00:ab:6e:

9c:e0:e0:13:b1:a3:41:d4:12:5f:3b:e0:e9:31:a3:c3:6e:d5:

02:21:00:c6:cf:1d:bd:9d:0f:b1:2c:5b:ef:17:4e:30:44:52:

c3:27:fa:a2:fa:40:29:f5:49:65:31:21:05:c2:bb:2b:35

Copyright © 2017 USB 3.0 Promoter Group. All rights reserved.

#### B.2.1.2.1. ACD

**Table B-1: Version TLV Fields**

| Offset | Field  | Size | Value         |
|--------|--------|------|---------------|
| 0      | Type   | 1    | 00h (VERSION) |
| 1      | Length | 1    | 02h           |
| 2      | Data   | 2    | 4000h         |

**Table B-2: XID TLV Fields**

| Offset | Field  | Size | Value     |
|--------|--------|------|-----------|
| 0      | Type   | 1    | 01h (XID) |
| 1      | Length | 1    | 04h       |
| 2      | Data   | 4    | 00001234h |

**Table B-3: Power Source Capabilities TLV Fields**

| Offset | Field  | Size | Value                                                                                                                                                                                                                                                                                                                                               |
|--------|--------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | Type   | 1    | 02h (POWER_SOURCE_CAPABILITIES)                                                                                                                                                                                                                                                                                                                     |
| 1      | Length | 1    | 16h                                                                                                                                                                                                                                                                                                                                                 |
| 2      | Data   | 22   | 02h (Version)<br>01h (FW Version)<br>01h (HW Version)<br>00h (Voltage Regulation)<br>03h (Hold Up time)<br>07h (Compliance)<br>01h (Touch Current)<br>00h (Reserved)<br>2A0Ah (Peak Current1)<br>2A0Ah (Peak Current2)<br>2A0Ah (Peak Current3)<br>00h (Touch Temp)<br>00h (Source Inputs)<br>00h (Batteries)<br>01h (Num PDOs)<br>2A01912Ch (PDOs) |

**Table B-4: Security Description TLV Fields**

| Offset | Field  | Size | Value                                                                                                                                 |
|--------|--------|------|---------------------------------------------------------------------------------------------------------------------------------------|
| 0      | Type   | 1    | 05h (SECURITY_DESCRIPTION)                                                                                                            |
| 1      | Length | 1    | 06h                                                                                                                                   |
| 2      | Data   | 6    | 00h (FIPS/ISO Identifier)<br>0000h (Common Criteria Identifier)<br>55h (Security Analysis Identifier)<br>1A0Ah (IC Vendor Identifier) |

**Table B-5: Playpen TLV Fields**

| Offset | Field  | Size | Value         |
|--------|--------|------|---------------|
| 0      | Type   | 1    | FDh (PLAYPEN) |
| 1      | Length | 1    | 04h           |
| 2      | Data   | 4    | 54455354h     |

**Table B-6: Vendor Extension TLV Fields**

| Offset | Field  | Size | Value                                 |
|--------|--------|------|---------------------------------------|
| 0      | Type   | 1    | FFh (VENDOR_EXTENSION)                |
| 1      | Length | 1    | 04h                                   |
| 2      | Data   | 4    | 1A0Ah (VID)<br>1234h (Vendor Defined) |

### B.2.2. Root Certificate

```
-----BEGIN CERTIFICATE-----
MIIBcDCCARagAwIBAgIBATAKBggqhkhjOPQQDAjAkMRIwEAYDVQQKDA1VU0ItSUYg
Q0ExDjAMBgNVBAMMBVVTQjo6MCIYDzE5NzAwMTAxMDAwMDAwWhgPOTk5OTEyMzEy
MzU5NTlaMCQxEjAQBgNVBAoMCMVVTQi1JRiBDQTEOMAwGA1UEAwwFVVNCOjowWTAT
BgcqhkhjOPQIBBgqhkhjOPQMBBwNCAAQWcYqkNu02Tci2l7m7go+whORaE+ziCWBe
6+YM85dWe4H5CtSHPCwetDJEzpl6m6O3+w/lNgIgAyIWddEj0sMVozUwMzAPBgNV
HRMBAf8EBTADAQH/MASGA1UdDwQEAwIBBjATBgNVHSUBAf8ECTAHBgVngREBATAK
BggqhkhjOPQQDAgNIADBFAiAQvuWGtCw81PjbuU0gY5SfldXWbITDulelaLfs7a+g
JwIhAMXGpvdj/E7y5ADM5ZIkIZv7C9dgqu22lMsn6NKa3puw
-----END CERTIFICATE-----
```

Certificate:

Data:

```
Version: 3 (0x2)
Serial Number: 1 (0x1)
Signature Algorithm: ecdsa-with-SHA256
Issuer: O=USB-IF CA, CN=USB::
Validity
Not Before: Jan 1 00:00:00 1970 GMT
Not After : Dec 31 23:59:59 9999 GMT
Subject: O=USB-IF CA, CN=USB::
Subject Public Key Info:
Public Key Algorithm: id-ecPublicKey
Public-Key: (256 bit)
pub:
04:16:71:8a:a4:36:ed:36:4d:c8:b6:97:b9:bb:82:
8f:b0:84:e4:5a:13:ec:e2:09:60:5e:eb:e6:0c:f3:
97:56:7b:81:f9:0a:d4:87:3d:cc:1e:b4:32:44:ce:
99:7a:9b:a3:b7:fb:0f:e5:36:02:20:03:22:16:75:
d1:23:d2:c3:15
ASN1 OID: prime256v1
NIST CURVE: P-256
X509v3 extensions:
X509v3 Basic Constraints: critical
CA:TRUE
```

Copyright © 2017 USB 3.0 Promoter Group. All rights reserved.



```
X509v3 Key Usage:
    Certificate Sign, CRL Sign
X509v3 Extended Key Usage: critical
    2.23.145.1.1
Signature Algorithm: ecdsa-with-SHA256
    30:45:02:20:10:be:e5:86:b4:2c:3c:d4:f8:db:b9:4d:20:63:
    94:9f:95:d5:d6:6c:84:c3:ba:57:a5:68:b7:ec:ed:af:a0:27:
    02:21:00:c5:c6:a6:f7:63:fc:4e:f2:e4:00:cc:e5:92:24:21:
    9b:fb:0b:d7:60:aa:ed:b6:d4:cb:27:e8:d2:9a:de:9b:b0
```

### B.2.3. Key Pairs

#### B.2.3.1. Root Key Pair

```
-----BEGIN EC PRIVATE KEY-----
MHcCAQEEIFQg8T7V348htdP6kyA5luN62FEbsZUsp+mNffff68gK5oAoGCCqGSM49
AwEHoUQDQgAEFnGKpDbtNk3Itpe5u4KPsITkWhPs4glgXuvmdPOXVnuB+QrUh3M
HrQyRM6Zepujt/sP5TYCIAMiFnXRI9LDFQ==
-----END EC PRIVATE KEY-----
Private-Key: (256 bit)
priv:
    54:20:f1:3e:d5:df:8f:21:b5:d3:fa:93:20:39:96:
    e3:7a:d8:51:1b:b1:95:2c:a7:e9:8d:7d:f7:fa:f2:
    02:b9
pub:
    04:16:71:8a:a4:36:ed:36:4d:c8:b6:97:b9:bb:82:
    8f:b0:84:e4:5a:13:ec:e2:09:60:5e:eb:e6:0c:f3:
    97:56:7b:81:f9:0a:d4:87:3d:cc:1e:b4:32:44:ce:
    99:7a:9b:a3:b7:fb:0f:e5:36:02:20:03:22:16:75:
    d1:23:d2:c3:15
ASN1 OID: prime256v1
NIST CURVE: P-256
```

#### B.2.3.2. Intermediate Key Pair

```
-----BEGIN EC PRIVATE KEY-----
MHcCAQEEINVK+bWvkojgcgKWe9uZvgldgZKxv3iWfkN3tj5oXA+doAoGCCqGSM49
AwEHoUQDQgAEPxERrJqhwkOOZthBwU+Hat81DqpzbIleX0IZs1z48zfInWTsZbEW
TcfmiWBiatXVrTch02A5OBsvAUtDCUyFsQ==
-----END EC PRIVATE KEY-----
Private-Key: (256 bit)
priv:
    00:d5:4a:f9:b5:af:92:88:e0:72:02:96:7b:db:99:
    be:09:5d:81:92:b1:bf:78:96:7e:43:77:b6:3e:68:
    5c:0f:9d
pub:
    04:3f:11:11:ac:9a:a1:c2:43:8e:66:d8:41:c1:4f:
    87:6a:df:35:0e:aa:73:6c:89:44:5f:42:19:b3:5c:
    f8:f3:37:c8:9d:64:ec:65:b1:16:4d:c7:e6:89:60:
    62:6a:d5:d5:ad:37:21:d3:60:39:38:1b:2f:01:4b:
    43:09:4c:85:b1
ASN1 OID: prime256v1
NIST CURVE: P-256
```

### B.2.3.3. Leaf Key Pair

```
-----BEGIN EC PRIVATE KEY-----
MHcCAQEEIGvMiauzTwqDQUNpuQZB/B6nqRA+bAwwc8Yd5umOxEKtoAoGCCqGSM49
AwEHoUQDQgAEqbD4ZrAtkSuHZCJX0q4LB+H6g6aOtE8/FnlDo+XYAHLcCtWzAKT7
i8AF0018nY1Iu6xGjOUot1scWvouTdo4RQ==
-----END EC PRIVATE KEY-----
Private-Key: (256 bit)
priv:
  6b:cc:89:ab:b3:4f:0a:83:41:43:69:b9:06:41:fc:
  1e:a7:a9:10:3e:6c:0c:30:73:c6:1d:e6:e9:8e:c4:
  42:ad
pub:
  04:a9:b0:f8:66:b0:2d:91:2b:87:64:22:57:d2:ae:
  0b:07:e1:fa:83:a6:8e:b4:4f:3f:16:79:43:a3:e5:
  d8:00:72:dc:0a:d5:b3:00:a4:fb:8b:c0:05:3b:4d:
  7c:9d:8d:48:bb:ac:46:8c:e5:28:b7:5b:1c:5a:fa:
  2e:4d:da:38:45
ASN1 OID: prime256v1
NIST CURVE: P-256
```

## B.3. Example Authentication Signature Verification

### B.3.1. CHALLENGE Request

```
01 83 00 00 ; CHALLENGE Request Header
46 29 65 BE EE 5B 63 45 B6 F6 31 72 A2 53 5A 35 ; Nonce
A3 D5 73 A4 45 F6 E0 3F B9 DB AA 43 FE DD A0 AF
```

### B.3.2. CHALLENGE\_AUTH Response

```
01 03 00 01 ; CHALLENGE_AUTH Response Header
01 01 01 00 ; Min/MaxProtocolVersion, Capabilities
66 09 26 B6 CB 61 86 5C 60 78 1A 98 92 AB F4 B7 ; CertChainHash
C2 4A B6 27 7C 2A 69 84 8A C6 90 B4 1C 18 63 E1
C2 6C 8B 49 AC 8C 52 48 4F C0 43 14 51 E4 98 F5 ; Salt
2D E5 32 0C DA B7 A0 B7 92 38 E6 44 54 2B 49 AC
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ; Context Hash
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
3C C4 AD FF 1B AC 89 14 4C F1 F5 6C AA 1A 96 CB ; Signature
DD C9 EC FD D5 AA 4F 7C E5 E5 91 D1 89 14 BF EB
99 FA D2 96 5C 09 B3 71 36 2E EF 8D 1C 6D 9D 87
07 A6 63 C6 5D 9B 8C 2F 25 D8 BE B1 83 B8 6F 63
```

## C Potential Attack Vectors

A list with examples of possible attacks against a Product is provided below. This list should be used as a checklist to determine whether a product has been thoroughly designed so that it can withstand known attacks that are most likely employed by common attackers. The list is partly based on the “Joint Interpretation Library - Application of Attack Potential to Smartcards” document and should be updated and reviewed regularly. A common criteria certificate with EAL5 and resistance against attackers with high attack potential determines that resistance against the attacks listed below has been achieved.

- Conformance testing of implemented algorithms (ECDSA, SHA256) according to test vectors and procedures described by NIST’s Cryptographic Algorithm Validation Program (CAVP). See <http://csrc.nist.gov/groups/STM/cavp/index.html>
- Protection of secret key operations against timing analysis.
- Protection (e.g., randomization/masking) against standard analysis of power consumption (SPA/DPA) and electromagnetic emanation (SEMA/DEMA) of secret key operations.
- Protection against advanced side-channel attacks against ECC-ECDSA computation (e.g., refined power analysis, zero value attacks, address-bit DPA, template attacks) of secret key operations.
- Protection critical computations against fault insertion (temperature, voltage, frequency variation; spikes and glitches; light; forcing; radiation) and advanced fault attacks (e.g., DFA, multi-bit faults).
- Protection of secret key in non-volatile memory against extraction (e.g., memory encryption) and manipulation/modification using non-invasive, semi-invasive, or invasive attacks.
- Protection against probing or forcing of intermediate values of the secret key during transfer on a chip internal bus.
- Protection and post-production lock down of test modes (e.g., JTAG) and scan chain.
- Protection against advanced invasive attacks like micro-probing or modification of circuits using a focused ion beam (FIB).
- Test of the statistical properties of the device-internal true random number generator (TRNG)
- Online test to recognize failure or manipulation by an attacker of the TRNG during operation.
- Secure system reset in case of the detection of an attack.
- Protection and thorough testing (e.g., fuzzing) to prevent (logical) attacks on software (e.g., bugs) like buffer overflows, man-in-the-middle, replay attacks, undocumented commands, bypass of authentication or access control.

---





INTERNATIONAL  
ELECTROTECHNICAL  
COMMISSION

3, rue de Varembé  
PO Box 131  
CH-1211 Geneva 20  
Switzerland

Tel: + 41 22 919 02 11  
Fax: + 41 22 919 03 00  
[info@iec.ch](mailto:info@iec.ch)  
[www.iec.ch](http://www.iec.ch)